SENTINEL-E: Anytime-Valid Sequential Detection for Streaming Visual Anomaly Surveillance

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Abstract

Streaming visual-anomaly systems — illegal dumping among them — are trained and evaluated as fixed-sample-size classifiers: given a short clip, emit one decision and report clip-level precision and recall. Deployment does not look like that: a municipal camera runs for weeks, and what an operator manages is the number of false alarms accumulated over the whole monitoring period, because each one costs a dispatch. Alerting the first time a per-frame threshold is crossed on a continuous stream is optional stopping, and it inflates the true false-alarm rate by an amount clip-level evaluation cannot show: a per-frame rule fed p-values individually valid at the 1% level raises at least one false alarm on 99.8% of 40-minute null streams. Our contribution is an algorithm. Every anytime-valid sequential change detector in use — CUSUM, Shiryaev–Roberts, e-detectors, non-partitioned e-detectors — models a switch that, once it happens, lasts forever. A dumping event is an *episode*: it begins, flickers under intermittent occlusion, ends, and recurs weeks later at the same site. We replace the mixture over changepoint locations with a mixture over episodes, realised as a two-state Markov chain whose forward recursion collapses a sum over 2^t trajectories into two numbers. The resulting *episodic e-process* is an exact non-negative supermartingale — so Ville's inequality gives time-uniform false-alarm control with no new machinery — costs $O(1)$ per frame against $O(t)$ for a general mixture over active starts, and contains the changepoint mixture exactly as the boundary case in which episodes never end, a strict generalisation along an axis the anytime-valid e-process family has left fixed at infinity. Around this core sits a per-camera conformal layer calibrated conditionally on observable context and corrected by exact Beta order-statistic levels, and a fleet layer in which a graph network steers each camera's episode hazard through a *predictable* modulation — provably unable to damage the guarantee — combined across cameras by e-BH under arbitrary dependence. Two evidence sources back these claims. Large-horizon simulated streams (10^5–10^6 frames) show every classical and parametric baseline we test — fixed threshold, per-frame p-value threshold, CUSUM, Shiryaev–Roberts, a parametric e-detector, E-SHIFT — failing to hold its nominal false-alarm level, while SENTINEL-E and the changepoint mixture it strictly contains both hold it; a paired comparison on a shared betting grid resolves a real advantage for the episodic mixture at two and four recurring episodes and a real, smaller cost at one. We also report where the guarantee stops holding: an unmodelled covariate shift beyond $\times 3.0$ inflates the rate unless a shift-weighting safeguard is engaged, at a steep cost in power. New in this revision: the pipeline is also run, unmodified, on real surveillance video (CUHK Avenue and UCSD Ped2) with a real optical-flow backbone rather than a simulator. This surfaces a limitation the simulated streams could not: real per-frame scores are autocorrelated across hundreds of frames (207 on Avenue, 357 on Ped2), close to or beyond a real episode's own duration, leaving only a narrow lag window in which validity and power coexist at all. Inside that window the guarantee holds on both real corpora — realised false-alarm rate 0.000 on Ped2 and 0.020 on Avenue at $\alpha = 0.1$ — while every classical baseline tested fails to hold it on the same real streams; detection power is real but modest on Ped2 and weak on Avenue, whose real episodes are shorter. This gap, not the simulated results, is the paper's central open problem. SENTINEL-E costs 0.61 µs per frame, 0.0016% of the detector it wraps.

Keywords: video anomaly detection; illegal dumping detection; anytime-valid inference; e-values;

conformal prediction; sequential change detection; false discovery rate; video surveillance; edge deployment

1 Introduction

Illegal dumping is a persistent municipal problem with an obvious computer-vision framing, and the field has adopted that framing consistently: collect clips, label them as containing a dumping event or not, train a video classifier, report clip-level accuracy. The 2026 IWDD contest [Bouwmans et al., 2026] gave the field a shared benchmark on these terms: ten teams trained on Mivia-IWDD (400 surveillance videos, balanced positive and negative, with onset timestamps) and were ranked on a private 100-video test set. Seven entries are published: X3D with temporal localisation [Almorsy and Torki, 2026], cross-fold ensembles of Hiera and VideoMAE [Kim et al., 2026], a three-headed classifier [Scarrica et al., 2026], a vision–language temporal analysis [Kairanbay, 2026], a temporal-spatial novelty detector [Zbuce et al., 2026], onset-centric boundary-sensitive estimation [Bhat and Lin, 2026], and a lightweight two-stage recurrent framework [Putzu et al., 2026]. Every one of them reports per-video precision, recall and F_1 together with a notification-delay measure, frame rate and peak memory, and the organisers' own overview additionally analyses the field in ROC space with an explicit false-positive-rate axis. Where "false alarms" appear in an individual entry, they are counted per clip — "six false alarms on thirty negative videos" [Almorsy and Torki, 2026] — which is specificity, not a rate over a monitoring period, and the same is true of the organisers' ROC analysis: it is a per-clip quantity, not one over a continuous horizon. The larger Mivia-IWDD-500 release [Greco et al., 2026] — 250 positive and 250 negative surveillance videos with onset timestamps, and the corpus this paper's protocol is built to run on — postdates the contest rather than having supplied it.

The framing does not match deployment. A camera watching a lay-by does not receive a curated ten-second clip and return a verdict; it produces frames continuously for weeks, and the system alerts the first time its evidence crosses a threshold. What the municipality manages is not per-clip precision but the number of enforcement dispatches wasted over a season. Those two quantities are separated by a step the clip-level protocol never exercises: *repeatedly* testing a stream and stopping at the first exceedance is optional stopping, and it invalidates the level at which any fixed threshold was calibrated.

The magnitude of the gap is easy to state and, we find, easy to underestimate. A rule that raises a false alarm on each frame with probability q raises at least one over T frames with probability $1 - (1 - q)^T$, which tends to one for any $q > 0$. Persistence filters and score smoothing change q; they do not change the limit. Feeding a per-frame rule p-values that are individually and provably valid at the 1% level — so that nothing is wrong with the calibration — produces at least one false alarm on 99.8% of 40-minute null streams (Figure 4 (a)); a raw score threshold tuned to a 10^{-3} per-frame rate reaches 0.640 within 0.2 hours and 1.000 by 2.7 hours (Figure 4 (b)). A clip-level evaluation cannot see this, because it never runs the detector for two hours.

Nothing in the argument above is specific to dumping. Any scene-level anomaly a per-frame backbone scores — loitering, wrong-direction motion, an abandoned object, a vehicle where pedestrians belong — exposes the same gap between a per-clip evaluation and a continuously monitored stream, and the method that follows is built and validated at that scope: an anytime-valid sequential detector for streaming visual anomaly surveillance, with illegal dumping as the motivating case throughout. We say this plainly rather than let it surface only in the data section: Section 4.2's empirical validation runs on two classical video-anomaly-detection corpora, not dumping footage, because no continuously-labelled real dumping stream of the required length is publicly available, and that scope mismatch is a real limitation this paper discloses (Section 5) rather than hides.

1.1 Why the existing sequential machinery is the wrong shape

The tools for optional stopping are mature. Ville's inequality [Ville, 1939] bounds the probability that a non-negative supermartingale *ever* exceeds a level, and the game-theoretic literature [Shafer and Vovk, 2019, Ramdas et al., 2023, Grünwald et al., 2024] turns that into a construction: bet against the null, and the accumulated wealth is a time-uniform certificate. Conformal prediction [Vovk et al., 2005] supplies distribution-free p-values to bet on, e-detectors [Shin et al., 2024] adapt the machinery to change detection, non-partitioned e-detectors [Saha and Ramdas, 2026] sharpen it to composite pre-change classes, and e-BH [Wang and Ramdas, 2022] combines evidence across streams under arbitrary dependence. To our knowledge none of it has been applied to waste, dumping or pollution surveillance.

Importing it unchanged would not be enough, and the reason is structural rather than a matter of tuning. *Every one of these procedures models a persistent change.* At an unknown frame ν the law switches from P_0 to P_1 and stays there for the rest of time; the mixtures are over ν alone, and the optimality theory, thresholds and delay bounds all presume it.

An illegal dumping event is not a persistent change. It is an *episode*. A van pulls up, the offender unloads for forty seconds and drives away. Inside the episode the signal is *intermittent*: the offender is occluded by the vehicle, steps out of frame, returns. Then the episode ends, the stream reverts to its null behaviour, and — because a site that has been used once is used again — another episode follows weeks later.

Modelling that as a persistent change is not a small mismatch. A changepoint-mixture wealth process keeps betting at full aggression after the episode is over, handing back what the episode earned; and because it treats every frame after ν as drawn from the post-change law, the quiet frames *inside* the episode count against it rather than being expected. Both effects lengthen delay, and the second worsens as intermittency increases — which is the regime this problem lives in. Empirically, over 8 recurring episodes per camera the changepoint mixture misses 0.010 of events where the episodic construction misses 0.000; at 4 episodes, a smaller but still substantial gap, it is 0.055 against 0.015 (Section 4.5).

1.2 Contributions

- **The episodic e-process** (Section 3.3), our main contribution. Mixing over episodes rather than changepoints, via a two-state Markov chain, gives a wealth process that is (i) an exact non-negative supermartingale, so Ville's inequality applies unchanged; (ii) computable in $O(1)$ time and memory per frame by an HMM forward recursion, where a general mixture over active starts costs $O(t)$ per observation [Saha and Ramdas, 2026]; and (iii) a strict generalisation, equal to the changepoint mixture exactly when the episode-end rate is zero (Proposition 4).

- **A quantity worth passing between cameras, whose one tested use did not pay off.** The same recursion returns the posterior probability that a camera is currently inside an episode. Unlike accumulated wealth it is bounded in $[0, 1]$ and comparable across cameras with different histories, which is the property a message-passing layer needs to aggregate it without one long-running camera swamping its neighbours — and Section 4.9 reports, as a negative result, that the one spatial-prior architecture we built on top of it does not shorten detection delay, so on the fleets tested here the graph layer's measured value comes entirely from e-BH at the combination step, not from this quantity. The property the posterior offers a future architecture is real; the specific use we made of it is not one of the paper's positive results.

- **Exact calibration-conditional conformal levels** (Theorem 1). Each attainable p-value level is replaced by its exact Beta order-statistic upper confidence bound rather than inflated additively. At $n = 2{,}000$, $\delta = 10^{-3}$ the smallest usable p-value becomes 0.0037 instead of the 0.0441 that Dvoretzky–Kiefer–Wolfowitz permits — a 11.9× improvement precisely where a betting detector earns its wealth, worth a factor of 8.3 in detection delay.

- **A learned spatial prior that provably cannot break the guarantee** (Theorem 5) — and the measurement showing it does not help either. Because the network's output enters only as a predictable modulation of the episode hazard and the betting stake, the wealth remains a supermartingale whatever it outputs. Our experiments confirm the theorem and refute the motivation: the learned controller leaves the false-alarm guarantee exactly intact while *reducing* power (Section 4.9). We report the construction because the invariance is worth having — it is what lets a learned component sit on top of a formal guarantee at all — and the result because it is what we found.

- **An evaluation that measures guarantees rather than asserting them**, over up to 1500 Monte-Carlo streams per operating point, against six baselines including a reimplementation of the closest generic prior art [Khan and Syed, 2026]; a component ablation in which 5 of 19 variants fail to deliver the level they claim; and a cost measurement putting the guarantee layer at 0.0016% of its backbone. We also correct a measurement pitfall that affects such comparisons generally: detection delay conditioned on detection is averaged over a different subset of runs for each method and systematically flatters the ones that miss the hard events, so we report a censored delay alongside it (Section 3.1).

- **A negative result we would have preferred not to have.** We introduced a dilated betting function to model within-episode intermittency, and the ablation shows the κ mixture already handles it: dilation costs delay and never repays it (Section 3.3.2). It is off by default and reported rather than buried.

What we do not claim. The changepoint mixture with retained tail mass that our construction reduces to is not new: it is the PFA-controlled e-detector of Saha and Ramdas [2026], which already delivers $\mathbb{P}(\tau < \infty) \leq \alpha$ for a mixture over start times with weights summing to one. Our contribution is the episode axis on top of it, and the $O(1)$ recursion that axis happens to make available. The conformal and e-BH layers are existing machinery, used as intended and credited as such; Section 5 states plainly what the evaluation does and does not establish.

2 Related work

Waste and dumping detection. Image-level litter detection is well served by TACO [Proença and Simões, 2020]; industrial waste-sorting imagery, a related but distinct acquisition regime of cluttered conveyor-belt scenes rather than in-the-wild litter, by ZeroWaste [Bashkirova et al., 2022]. UAV and wide-area monitoring is served by UAVVaste [Kraft et al., 2021] and AerialWaste [Torres and Fraternali, 2023]; marine pollution by MARIDA [Kikaki et al., 2022] and MADOS [Kikaki et al., 2024].

Video-based dumping detection now has a public benchmark and an active contest around it. The 2026 IWDD contest [Bouwmans et al., 2026] drew ten teams, trained on the released Mivia-IWDD set (400 videos, 200 positive/200 negative) and ranked on a private 100-video test set by precision, recall and F1 plus normalised notification delay, processing frame rate and peak memory; the larger Mivia-IWDD-500 release we build our protocol on [Greco et al., 2026] postdates the contest rather than having supplied it. Seven of the ten entries are published: 3D CNNs with temporal localisation [Almorsy and Torki, 2026], cross-fold transformer ensembles [Kim et al., 2026], multi-head classifiers [Scarrica et al., 2026], vision–language temporal reasoning [Kairanbay, 2026], novelty detection [Zbuce et al., 2026], onset-centric boundary-sensitive estimation [Bhat and Lin, 2026], and a lightweight two-stage recurrent framework [Putzu et al., 2026]. These are strong backbones and our work does not compete with them: SENTINEL-E consumes a frozen backbone's scores, so a better backbone and this work compound. What none of them reports — and what the clip-level protocol they share cannot express — is a false-alarm rate over a continuous monitoring horizon, an average run length, or a detection delay under continuous monitoring; the contest overview does analyse the field in ROC space with an explicit false-positive-rate axis, which is still a per-clip quantity rather than a rate over a horizon, so the distinction we draw survives with that qualification attached.

Video anomaly detection benchmarks (2025–2026). The classical video anomaly detection (VAD) literature this paper's real-data protocol draws its streams from — CUHK Avenue [Lu et al., 2013] and UCSD Ped2 [Mahadevan et al., 2010] — continues to report clip- or frame-level ROC-AUC as its primary metric: a reconstruction-based idempotent generative network reaches 99.03% AUC on Ped2 and 92.40% on Avenue [Dong et al., 2025]. That method post-dates this paper's original submission and outperforms, on frame-level AUC alone, the Farneback optical-flow backbone Section 4.2 uses (0.763/0.783). We do not run it head to head against SENTINEL-E, and the reason is the point this paper makes rather than an omission: AUC is computed after the fact over a fixed, already-labelled test set and says nothing about the rate at which a rule raises an alarm on a continuously monitored, unlabelled stream, which is the quantity Section 4.2 actually measures. Rashidi [2026] audits this gap directly: eleven off-the-shelf VAD feature sets that average 70.4% AUC on their own training distribution collapse to 49.9% — chance — under a cross-dataset shift, with false-alarm rates around 3.2×10^4 per hour at thresholds a same-dataset evaluation would call operational. That finding and Section 4.2's own (the narrow, dataset-specific lag window a time-uniform guarantee actually needs on real footage) are independent audits of the same failure mode from opposite ends: a benchmark-AUC number does not certify what happens when the clock keeps running.

Sequential detection of changes that end. Classical quickest-detection theory treats a permanent shift, but a sub-literature treats exactly the object this paper does: a change of finite, possibly unknown duration. Transient change detection [Guépié et al., 2012a, 2017] minimises worst-case missed detection under a false-alarm constraint for a change that must be caught before it disappears, using a window-limited CUSUM; the intermittent-change line [Sokolov et al., 2023] treats a change of *unknown* duration directly, and a two-state hidden Markov model is used for quickest detection in Fuh and Mei [2015] — structurally the closest classical relative of the two-state chain in Equation (3). Epidemic changepoint detection [Fisch et al., 2022, Juodakis and Marsland, 2023] treats a related but distinct problem, a segment that departs from and then returns to a background level estimated from the same series, going back to the original two-changepoint epidemic-alternative formulation. None of this literature is ARL- or Ville-type time-uniform: it is CUSUM-, SPRT- or window-shaped with asymptotic or fixed-sample-size guarantees, not an e-process. What we add is not the finite-duration change itself but an anytime-valid, conformal-fed construction of it; Section 3.3 states that scope explicitly rather than claiming the axis itself is new.

On the construction: mixing a bounded-wealth process over the trajectories of a two-state chain and collapsing them by a forward pass is an instance of the fixed-share, or switching-expert, family [Herbster and Warmuth, 1998, van Erven et al., 2012], of which Equation (3) is a two-expert case with one expert held at the neutral bet. What that literature does not supply is the martingale property under optional stopping that Theorem 2 establishes; conversely, what we do not claim is that mixing over a switching structure is itself new.

Closer to our architecture than either of these are the conformal test martingales of Vovk et al. [2003] and their successors, which combine a conformal front end with a Ville-thresholded martingale for online change and exchangeability testing — the same two-layer shape as Sections 3.2 and 3.3. We found no instance of that combination, nor of any other e-process or anytime-valid conformal machinery, applied to waste, dumping, or pollution surveillance specifically, though classical sequential change detection has reached adjacent environmental problems, including contamination detection in water-distribution networks [Guépié et al., 2012b].

Sequential change detection. CUSUM [Page, 1954], Shiryaev's Bayesian statistic [Shiryaev, 1963] and Shiryaev–Roberts [Roberts, 1966] are the classical procedures, with optimality theory due to Lorden [1971] and Moustakides [1986]. Their guarantees are asymptotic average-run-length statements presuming a correctly specified pre-change law; detector scores from real footage are autocorrelated, heavy-tailed and diurnally non-stationary, so the presumption fails and both lose control in our experiments. Shin et al. [2024] give a nonparametric e-detector framework with non-asymptotic ARL bounds, and Saha and Ramdas [2026] extend it to composite pre-change classes, including a PFA-controlled variant that mixes over start times with weights summing to one and thereby achieves $\mathbb{P}(\tau < \infty) \leq \alpha$. That variant is exactly what our Proposition 4 reduces to, and we claim no credit for it.

The difference is the change model. All of these mix over the changepoint *location* of a persistent shift, and none of them is an e-process: what they lack is not the finite-duration axis itself, which classical quickest-detection theory has treated since the transient- and intermittent-change literature discussed above, but a time-uniform, optional-stopping-safe construction of it that stays $O(1)$ per frame and feeds from a conformal front end rather than a parametric one. That is the axis Section 3.3 adds *to this family.* A secondary but practical difference is cost. Evaluated exactly across all active starts, a non-partitioned e-detector is $O(t)$ per observation [Saha and Ramdas, 2026]; under a geometric prior over start times the sum telescopes to an $O(1)$ recursion, which is the special case of Equation (4) at $\eta = 0$. The point is that the episodic process retains that $O(1)$ cost *while adding the episode-end axis to the e-process family specifically* — the extra generality is free arithmetically as well as statistically, which matters when the horizon is a month of video and the hardware is a camera.

Anytime-valid inference and e-values. Testing by betting [Shafer, 2021, Ramdas et al., 2023, Grünwald et al., 2024] turns Ville's inequality into a design principle; Waudby-Smith and Ramdas [2024] develop the betting strategies we adapt. Wang and Ramdas [2022] introduce e-BH, which controls FDR under arbitrary dependence and underpins our fleet layer; Vovk and Wang [2021] characterise e-value merging. Closest to this work in spirit is E-SHIFT [Khan and Syed, 2026], which detects distribution shift in streaming learning systems with a sub-exponential e-process built on a composite non-conformity score, and which already provides an α-mixing extension for temporally correlated streams. We reimplement it as a baseline. It differs from SENTINEL-E in three ways that matter here: its log-moment-generating-function bound is estimated, so by the authors' own account validity is asymptotic in the calibration size rather than finite-sample; it assumes i.i.d. calibration and deployment scores, which is stronger than the exchangeability our conformal layer needs and which a diurnal cycle breaks regardless; and its wealth starts at frame one after each alert, so evidence arriving hours into a stream must first repay what was burned before it, exactly as our own restart convention does between alarms.

Conformal prediction. Split conformal prediction [Vovk et al., 2005] supplies the distribution-free p-values we bet on. Vovk [2012] studies calibration-conditional validity, which is the concern our Beta levels address, and Tibshirani et al. [2019] handle covariate shift by likelihood-ratio weighting, which we implement and evaluate. Conformalising a normalised regression residual is standard practice [Lei et al., 2018, Romano et al., 2019], and localisation [Guan, 2023] pursues conditional coverage by other means. Our Layer 1 combines these with the sequential requirement, which imposes a constraint the prediction-interval literature does not face: the p-values must be super-uniform *conditionally on the past*, not merely marginally, or the wealth process is not a supermartingale.

3 Method

3.1 Problem formulation

A camera produces frames $x_1, x_2, \ldots$ and a frozen detector g produces scores $s_t = g(x_t) \in \mathbb{R}$, with higher meaning more dumping-like. Alongside each frame we observe a low-dimensional context vector $c_t \in \mathbb{R}^d$ — time of day, a weather flag, ambient luminance, a background-subtraction motion measure — available to any deployed camera without privileged knowledge. The operator supplies a per-camera calibration set $\mathcal{D} = \{(s_i^{\text{cal}}, c_i^{\text{cal}})\}_{i=1}^n$ of confirmed no-dumping frames.

Write $\mathbb{P}_\infty$ for the law of the stream under "no dumping ever" and $\mathbb{P}_\nu$ for the law when dumping begins at frame ν. A detection rule is a stopping time τ with respect to the observation filtration $(\mathcal{F}_t)$. We require:

Definition 1 (Time-uniform false-alarm control)**.** A rule has time-uniform false-alarm level α if $\mathbb{P}_\infty(\tau < \infty) \leq \alpha$.

The quantifier is the whole point. A per-frame guarantee $\mathbb{P}_\infty(\text{alarm at } t) \leq \alpha$ for each fixed t says nothing about $\mathbb{P}_\infty(\exists t : \text{alarm at } t)$, and the gap between them grows with the horizon. Definition 1 is what a municipality is implicitly buying when it is told a detector has a 1% false-alarm rate.

Dumping is episodic, so we write $\mathbb{P}_\mathcal{E}$ for the law when a set of episodes $\mathcal{E} = \{[\nu_1, \nu_1 + L_1], \ldots\}$ occurs, within each of which a fraction π of frames are anomalous. The single-persistent-change model is the special case of one episode with $L = \infty$ and $\pi = 1$.

We report, over Monte-Carlo replicates: PFA, the probability that a null stream of a stated length raises at least one alarm; ARL_0, the mean frames to a false alarm on a null stream, censored at the horizon; ADD, the mean frames from the first onset ν_1 to the first alarm at or after it, over runs that alarm; and the miss rate, the fraction of post-change segments ending unflagged.

A measurement pitfall, and the fix. Conditional ADD is averaged over a *different subset of runs for each method*: a detector that misses the hard events is scored only on the easy ones and therefore looks fast. Since the methods compared here differ substantially in miss rate, we also report a *censored* delay in which a run that never alarms is charged the whole remaining horizon rather than dropped. The two can rank methods oppositely — in Section 4.5 the changepoint mixture has the shorter conditional delay and the longer censored delay — and where they disagree the censored figure is the comparable one, so head-to-head claims in this paper rest on it. This is not specific to our setting and we suspect it affects published comparisons elsewhere.

Every comparison is made at a method's *realised* PFA, never its nominal one, since a method whose nominal level is meaningless would otherwise appear to dominate by misreporting its own error rate.

From clips to streams. A clip-labelled corpus such as Mivia-IWDD becomes a streaming benchmark without new data collection. Reserve a disjoint subset of confirmed-negative clips as $\mathcal{D}$; concatenate the remaining negative clips into a background stream of the desired length; splice positive clips in at random onsets. Holding the calibration clips out preserves the exchangeability the conformal layer needs. This protocol is implemented in the library (`build_stream_from_clips`) and is how Section 4.2's Ped2 and Avenue streams are built once per-frame backbone scores for a real corpus are in hand.

3.2 Layer 1: conformal calibration

Betting rate. The e-process of Section 3.3 requires increments that are conditionally super-uniform given the past. Video violates this badly: at 25 fps the lag-one autocorrelation of backbone scores exceeds 0.85 in our streams. We estimate a decorrelation lag ℓ from the calibration set alone — projecting out the context-predictable drift first with a quadratic least-squares fit, since otherwise the slow diurnal component dominates the autocorrelation and returns a uselessly large ℓ — and place one bet per ℓ frames. In our streams $\ell = 18$, so bets occur more than once per second, far finer than the timescale of a dumping event. The same reasoning applies to the calibration frames, which are consecutive video too: they are thinned by ℓ before use, so the honest calibration size is n/ℓ and it is that number the guarantee is stated in. Section 4.12 shows this is not a refinement: betting on every frame is the single largest source of invalidity we measured.

Conditional calibration. Let S be a nonconformity score. The split-conformal p-value of a deployment frame is $p = (1+|\{i : S_i \geq S\}|)/(n+1)$, which is super-uniform under exchangeability. Exchangeability between a night-time frame and a daylight-dominated $\mathcal{D}$ is exactly what fails. We therefore conformalise the *context-normalised residual*

$$z \;=\; \frac{s - \hat{g}(c)}{\hat{\sigma}(c)}, \qquad \hat{g}(c) \approx \mathbb{E}[s \mid c], \quad \hat{\sigma}(c) \approx \mathbb{E}\big[|s - \hat{g}(c)| \,\big|\, c\big], \tag{1}$$

with $\hat{g}$ and $\hat{\sigma}$ ridge regressions on a quadratic-with-interactions basis in c, fitted on one half of $\mathcal{D}$ and conformalised on the other. The split matters: it makes the conformalisation residuals independent of the fitted models, so conditionally on those models Theorem 1 applies verbatim. What remains assumed is that the normalised residual in Equation (1) is distributed independently of c — exchangeability is bought by the normalisation, not by the split — and Section 5 returns to what happens when the model is wrong. The gain over binning is that the whole held-out half backs every context value, rather than one bin's worth backing each. We implement Mondrian calibration as well, since it is exactly group-conditional and preferable when a single context variable dominates; Section 4.8 shows where each wins.

Because a regression can be extrapolated but a guarantee cannot, frames whose context falls outside the region $\mathcal{D}$ covers — a per-coordinate box widened by a margin, intersected with a Mahalanobis check for jointly novel combinations — are reported inactive, and the detector bets neutrally on them. A regime that was never calibrated cannot be certified, and the abstention rate reports how often that happens rather than hiding it.

Calibration-conditional validity. The same finite $\mathcal{D}$ scores every frame for months, so the relevant statement is conditional on $\mathcal{D}$, and conditionally the p-values deviate from uniform by $O(n^{-1/2})$. The standard remedy adds the Dvoretzky–Kiefer–Wolfowitz deviation $\varepsilon_n(\delta) = \sqrt{\log(2/\delta)/(2n)}$ [Dvoretzky et al., 1956, Massart, 1990] to every p-value. It works and it is far too blunt: at $n = 2{,}000$, $\delta = 10^{-3}$ it moves the smallest attainable p-value from 1/2001 to 0.0441, capping the evidence a single frame can supply at well under one nat. The tail is the only part a betting detector uses.

We use the exact distribution instead.

Theorem 1 (Exact calibration-conditional levels)**.** *Let $S_1, \ldots, S_n$ be i.i.d. from a continuous F and let S be an independent draw from F. For $j = 0, \ldots, n-1$ let $\delta_j > 0$ with $\sum_j \delta_j \leq \delta$, and set*

$$q_j \;=\; \mathrm{Beta}^{-1}(1 - \delta_j;\; j+1,\; n-j), \qquad q_n = 1. \tag{2}$$

Define $\tilde{p} = q_J$ with $J = |\{i : S_i \geq S\}|$. Then there is an event E with $\mathbb{P}(E) \geq 1 - \delta$, measurable with respect to $S_1, \ldots, S_n$, on which $\mathbb{P}(\tilde{p} \leq u \mid S_1, \ldots, S_n) \leq u$ for every $u \in [0, 1]$.

Proof. Conditionally on the calibration sample, $\{J \leq j\} = \{S > S_{(n-j)}\}$ where $S_{(1)} \leq \cdots \leq S_{(n)}$ are the order statistics, so $\mathbb{P}(J \leq j \mid \mathcal{D}) = 1 - F(S_{(n-j)})$. Since F is continuous, $F(S_{(k)}) \sim \mathrm{Beta}(k, n+1-k)$, hence $1 - F(S_{(n-j)}) \sim \mathrm{Beta}(j+1, n-j)$. By Equation (2), $\mathbb{P}(1 - F(S_{(n-j)}) >$

$q_j) = \delta_j$. Let E be the intersection over j of the complements; a union bound gives $\mathbb{P}(E) \geq 1 - \sum_j \delta_j \geq 1 - \delta$. On E, $\mathbb{P}(\tilde{p} \leq q_j \mid \mathcal{D}) = \mathbb{P}(J \leq j \mid \mathcal{D}) \leq q_j$ for every j, and since $\tilde{p}$ takes values only in $\{q_0, \ldots, q_n\}$ and q_j is non-decreasing in j, super-uniformity at the atoms extends to all u. $\square$

We spend the budget as $\delta_j \propto (j+1)^{-2}$, concentrating confidence where the betting functions look; under this allocation $\{q_j\}$ is verified non-decreasing in j, which is what identifies $\{\tilde{p} \leq q_j\} = \{J \leq j\}$ in the proof above (the implementation takes a running maximum as a safeguard against an allocation for which this fails). At $n = 2{,}000$, $\delta = 10^{-3}$ this gives $q_0 = 0.0037$ against DKW's 0.0441. Theorem 1 assumes i.i.d. calibration draws, which is why the calibration set is thinned; Section 4.12 does not resolve whether skipping the thinning costs validity at the replication count run there (Section 5), though the calibration-conditional argument no longer applies once consecutive draws are dependent.

Finally, when deployment reweights context that $\mathcal{D}$ does cover, weighted conformal calibration [Tibshirani et al., 2019] applies. We implement it with a logistic density-ratio estimate, gate it behind a held-out classifier two-sample test so it engages only when a shift is detectable, and take the calibration-conditional correction at the Kish effective sample size [Kish, 1965] in place of n in Equation (2). Unlike Theorem 1 itself, this substitution is not accompanied by a proof: Theorem 1's order-statistic argument assumes equal weights, and the Kish size is an approximation to the effective count under unequal weights that is exact only in that equal-weight limit, not a derived calibration-conditional bound for the weighted case. Section 4.8 reports that in our setting the residual construction of Equation (1) dominates it on power, which we take to be a fact about the setting rather than about the technique: reweighting corrects the marginal context distribution, while a sequential test is broken by *conditional* miscalibration; Section 5 returns to what the weighted variant's validity evidence does and does not show.

3.3 Layer 2: the episodic e-process

This is the core of the method. Let $p_1, p_2, \ldots$ be the p-values at the betting instants. A *betting function* is a non-increasing $f : (0,1] \to [0, \infty)$ with $\int_0^1 f \leq 1$; then $\mathbb{E}_{H_0}[f(p)] \leq 1$ for any super-uniform p, so $f(p)$ is an e-value. We use the power family $f_\kappa(p) = \kappa p^{\kappa-1}$, $\kappa \in (0,1]$, with $\kappa = 1$ the neutral bet $f \equiv 1$.

3.3.1 Mixing over episodes

Let $z_t \in \{\text{quiet}, \text{active}\}$ be a two-state Markov chain with onset probability $\mathbb{P}(\text{quiet} \to \text{active}) = \rho$ and *end* probability $\mathbb{P}(\text{active} \to \text{quiet}) = \eta$, started quiet, and let the wealth mix over every trajectory of that chain:

$$W_t \;=\; \sum_{z_{1:t}} \mathbb{P}(z_{1:t}) \prod_{s \leq t} f(p_s)^{\mathbf{1}\{z_s = \text{active}\}}. \tag{3}$$

Each trajectory contributes the product of betting factors over the frames it declares active, weighted by its prior probability. The classical mixture over changepoints is the special case in which the chain, once active, can never return: it mixes over *when* a change starts, whereas Equation (3) mixes over *when it starts and when it stops*, and over how many times that happens.

The sum has 2^t terms and collapses to two numbers. Writing A_t and Q_t for the parts of the sum whose trajectories end active and quiet respectively,

$$A_t = \big[(1-\eta)A_{t-1} + \rho\, Q_{t-1}\big] f(p_t), \qquad Q_t = \eta\, A_{t-1} + (1-\rho)\, Q_{t-1}, \tag{4}$$

with $A_0 = 0$, $Q_0 = 1$ and $W_t = A_t + Q_t$. This is the forward recursion of a hidden Markov model, and it is $O(1)$ per frame in both time and memory.

Theorem 2 (Time-uniform validity of the episodic e-process). *Under $\mathbb{P}_\infty$, if the p_t are conditionally super-uniform given $\mathcal{F}_{t-1}$, then (W_t) defined by Equation* (3) *is a non-negative supermartingale with $W_0 = 1$, and for any $\alpha \in (0, 1)$*

$$\mathbb{P}_\infty(\exists\, t \geq 1 : W_t \geq 1/\alpha) \;\leq\; \alpha.$$

Proof. Non-negativity is immediate and $W_0 = Q_0 = 1$. From Equation (4), A_t carries exactly one factor $f(p_t)$ and Q_t carries none, so

$$\begin{aligned}\mathbb{E}[W_t \mid \mathcal{F}_{t-1}] &= \big[(1-\eta)A_{t-1} + \rho Q_{t-1}\big]\, \mathbb{E}[f(p_t) \mid \mathcal{F}_{t-1}] \;+\; \eta A_{t-1} + (1-\rho)Q_{t-1} \\ &\leq (1-\eta)A_{t-1} + \rho Q_{t-1} + \eta A_{t-1} + (1-\rho)Q_{t-1} \;=\; A_{t-1} + Q_{t-1} \;=\; W_{t-1},\end{aligned}$$

using $\mathbb{E}[f(p_t) \mid \mathcal{F}_{t-1}] \leq 1$ and the fact that the transition probabilities out of each state sum to one. Ville's inequality [Ville, 1939] applied to the non-negative supermartingale (W_t) started at one gives the bound. □

Corollary 3 (End-to-end level). *Let E be the calibration event of Theorem 1, with $\mathbb{P}(E) \geq 1 - \delta$, on which the deployment p-values $\tilde{p}_t$ are conditionally super-uniform given $\mathcal{F}_{t-1}$. Then the episodic e-process run on $(\tilde{p}_t)$ satisfies $\mathbb{P}_\infty(\exists\, t \geq 1 : W_t \geq 1/\alpha) \leq \alpha + \delta$.*

Proof. On E, Theorem 2 applies verbatim and gives probability at most α conditional on $\mathcal{D}$; off E, the probability is bounded by 1, and $\mathbb{P}(E^c) \leq \delta$. Removing the conditioning, $\mathbb{P}_\infty(\exists t : W_t \geq 1/\alpha) \leq \mathbb{P}(E) \cdot \alpha + \mathbb{P}(E^c) \cdot 1 \leq \alpha + \delta$. □

This is the guarantee actually delivered end to end, and it is what the library documents (`sentinel_e/pipeline.py`) even where earlier drafts of this text quoted α alone. With the default $\delta = 10^{-3}$ this is a 10–20% relative loosening at the operating points Section 4.4 reports ($\alpha \in \{0.005, \ldots, 0.5\}$), not a rounding matter, and every level plotted in this paper should be read as $\alpha + \delta$.

Two things are worth drawing out. First, the episode structure is *free*: it needed no new machinery and cost nothing in the guarantee, because the transition probabilities telescope exactly. Second, the recursion never grows. Evaluated term by term, a mixture over candidate starts carries one term per start, at $O(t)$ work per observation [Saha and Ramdas, 2026]; Equation (4) carries two numbers per grid point whatever t is. The geometric-prior changepoint mixture also telescopes to $O(1)$ (Proposition 4), so the honest statement is not that we are cheaper than that special case but that we match it while spanning episodes of finite duration — month-long monitoring on edge hardware stays arithmetically trivial (Section 4.10).

Proposition 4 (Strict generalisation). *With $\eta = 0$ the recursion Equation* (4) *coincides with the mixture over changepoint locations under a geometric prior with hazard ρ; discarding Q_t from that recovers in turn the classical Shiryaev–Roberts statistic.*

Proof. With $\eta = 0$, $Q_t = (1-\rho)Q_{t-1}$ so $Q_t = (1-\rho)^t$ is the prior mass on "no episode has begun by t", and $A_t = (A_{t-1} + \rho Q_{t-1}) f(p_t)$ with $\rho Q_{t-1} = w_t$ the geometric prior mass on a change at t. Unrolling gives $A_t = \sum_{j \leq t} w_j \prod_{s=j}^t f(p_s)$, whence $W_t = \sum_{j \leq t} w_j \prod_{s=j}^t f(p_s) + \sum_{j>t} w_j$, Shiryaev's Bayesian statistic under the geometric prior with hazard ρ [Shiryaev, 1963]. Dropping Q_t leaves $\sum_{j \leq t} w_j \prod_{s=j}^t f(p_s)$ alone, which is not a supermartingale — the telescoping in Theorem 2 fails by exactly w_t per step — and, taking the $\rho \to 0$ limit under the appropriate rescaling, recovers the classical Shiryaev–Roberts sum with unit start weights [Roberts, 1966]; at fixed $\rho > 0$ the two named statistics remain distinct objects, and only the proof's algebraic point — that discarding the quiet mass breaks the martingale property — depends on neither. □

So the episodic process sits strictly above both, and the axis it adds — episode duration — is one this anytime-valid e-process family has held fixed at infinity; Section 2 positions that claim against the classical transient- and intermittent-change literature, which has not. The empirical consequence is in Section 4.5: where the classical model is right, the two coincide within the noise of a paired comparison; where events recur, a gap opens whose size that same section resolves with uncertainty rather than by a single point estimate.

3.3.2 Intermittency, by dilation — a negative result

Within an episode only a fraction π of frames show the act, which suggests *dilating* the emission:

$$f_\pi(p) \;=\; (1-\pi) + \pi\, f(p). \tag{5}$$

This is legal, and cheaply so: f_π is a convex combination of the neutral bet and f, hence non-increasing with $\int_0^1 f_\pi \le 1$, so it is a betting function and Theorem 2 applies verbatim. It also says the right thing — *this frame belongs to an episode, but it may or may not show the act.*

We expected this to be the answer to intermittency. It is not, and we report that rather than quietly shipping it. Mixing over κ already handles intermittency: a stream in which a third of the episode frames are anomalous is best played with a milder bet, and the κ grid contains one. A π axis therefore buys no behaviour the grid did not already have, while spending prior mass to do it. In the ablation, enabling it lengthens censored detection delay from 244.5 s to 269.8 s (Table 18), and we found no intermittency level at which it repaid that cost — below $\pi \approx 0.2$ both variants fail together, and above it the plain κ mixture is ahead. The default grid therefore sets $\pi = 1$.

We keep Equation (5) in the exposition and the implementation for two reasons. It is the natural thing to try, so a reader is owed the evidence rather than left to repeat the experiment; and the construction remains available for a backbone whose anomalous frames are far stronger relative to its quiet ones than ours, where the trade-off could plausibly reverse. What it is not is a contribution, and we do not count it as one.

3.3.3 Staying parameter-free

Nothing above requires knowing $(\rho, \eta, \pi, \kappa)$. We place a prior grid over all four and run one recursion per grid point, mixing the results: $W_t = \sum_g \lambda_g W_t^{(g)}$ with $\sum_g \lambda_g = 1$ is a convex combination of supermartingales and hence a supermartingale. The deployed grid is 6 values of κ by 4 of η by 1 of π (dilation off by default, Section 3.3.2), i.e. 24 points, so the per-frame cost is a few dozen floating-point operations and still $O(1)$ in t; enabling the π grid (Section 4.12) triples the point count and the cost with it, still $O(1)$. In practice the mixture is what makes the method usable: an operator does not know how long a dumping event lasts at their site, and the grid spans everything from a ten-second burst to a permanent change.

3.3.4 The episode posterior

The recursion yields, at no extra cost, the posterior probability that the camera is inside an episode right now,

$$\Pi_t \;=\; \frac{A_t}{A_t + Q_t} \;\in\; [0,1], \tag{6}$$

prior-averaged over the grid. This is a more useful thing to look at, and to transmit, than accumulated wealth: it is bounded, it is interpretable as a probability, and it is comparable between a camera installed last week and one that has been running for a month. Section 3.4 uses it as the message the camera graph exchanges.

3.3.5 Numerics

Carrying Equation (4) in log space would need two `logaddexp` calls per grid point per frame. Instead we renormalise (A_t, Q_t) to sum to one at every step and accumulate the logarithm of the

Algorithm 1 One frame of the episodic e-process, vectorised over the prior grid

Require: prior grid $\{(\kappa_g, \eta_g, \pi_g)\}_{g=1}^{G}$ with log-prior weights $\log \lambda_g$, $\sum_g \lambda_g = 1$; onset hazard ρ (or ρ_t, if predictable modulation is enabled, Theorem 5); alarm threshold $1/\alpha$

1: $A_g \leftarrow 0,\ Q_g \leftarrow 1,\ s_g \leftarrow 0\ \forall g$ ▷ s_g: per-grid-point log-scale; $A_g + Q_g = 1$ is the invariant
2: **for** each betting instant $t = 1, 2, \ldots$ **do**
3: receive conformal p-value p_t (or skip this t: thinning / abstention)
4: **for** each grid point g **(vectorised, not a real loop) do**
5: $f_g \leftarrow (1 - \pi_g) + \pi_g\, \kappa_g\, p_t^{\kappa_g - 1}$ ▷ dilated power bet, Equation (5); $\pi_g = 1$ if dilation off
6: $A'_g \leftarrow \big[(1 - \eta_g)\, A_g + \rho\, Q_g\big] f_g$
7: $Q'_g \leftarrow \eta_g\, A_g + (1 - \rho)\, Q_g$ ▷ Equation (4), one grid point
8: $c_g \leftarrow A'_g + Q'_g$ ▷ restores the invariant; $c_g > 0$ always
9: $A_g \leftarrow A'_g / c_g,\ \ Q_g \leftarrow Q'_g / c_g$ ▷ HMM scaling trick: the two-state simplex cannot underflow
10: $s_g \leftarrow s_g + \log c_g$
11: **end for**
12: $\log W_t \leftarrow \operatorname{logsumexp}_g\big(\log \lambda_g + s_g\big)$ ▷ mixture happens here, in log space, not by renormalising a joint sum
13: $\Pi_t \leftarrow \sum_g \lambda_g\, e^{s_g} A_g \,/\, e^{\log W_t}$ ▷ episode posterior, Equation (6)
14: **if** $\log W_t \geq \log(1/\alpha)$ **then**
15: **alarm**; if restarting, reset $A_g \leftarrow 0$, $Q_g \leftarrow 1$, $s_g \leftarrow 0\ \ \forall g$
16: **end if**
17: **end for**

scale factor, which is the standard HMM scaling trick: the two-state simplex cannot underflow, $\log W_t$ is the accumulated log-scale, and the per-step work is a handful of vector operations on the grid. Algorithm 1 states the per-frame update in full; it is a direct transcription of `sentinel_e/episodic.py`'s inner loop, not a simplification of it.

Two things about Algorithm 1 are worth making explicit because they are easy to miss on a first read of Equation (4) alone. First, every grid point keeps its *own* running invariant $A_g + Q_g = 1$ and its own log-scale s_g; the mixture over the grid is not a joint renormalisation but a single `logsumexp` taken only when $\log W_t$ or Π_t is read out, so adding grid points costs a longer vector, not a different algorithm. Second, the inner loop over g is the only place the grid appears and it is fully vectorised in the implementation (one array operation per line, not a Python loop), so nothing about the recursion's cost depends on how the grid was chosen: the parameter-free stance of Section 3.3.3 costs exactly $|\{g\}|$ times the single-hypothesis cost, not more. The normalising division (line 8) is exact, not a truncation or a clip, so no approximation enters between Equation (4) and what the code computes beyond ordinary floating-point rounding, which Section 4.4's oracle-p-value arm is designed to expose if it mattered.

A worked numeric trace. Table 1 runs four steps of Algorithm 1 by hand, at a single grid point ($\kappa = 0.3$, $\eta = 0.1$, $\rho = 0.01$, dilation off) so that every entry is checkable with a calculator: $f_t = \kappa\, p_t^{\kappa - 1}$, $A'_t = [(1 - \eta) A_{t-1} + \rho\, Q_{t-1}]\, f_t$, $Q'_t = \eta\, A_{t-1} + (1 - \rho)\, Q_{t-1}$, $c_t = A'_t + Q'_t$, and the running wealth is $W_t = \exp(\sum_{s \leq t} \log c_s)$.

Three things this trace makes concrete that the general statement can leave abstract. First, a single small p-value ($t = 1$, $p_1 = 0.02$, a moderately surprising frame) moves A_t from 0 to 0.045 in one step but moves W_t by less than 4% — the betting factor $f_1 = 4.64$ is diluted by the prior mass $Q_0 = 1$ still sitting on "no episode yet", which is the entire content of mixing over onset times rather than betting on frame one being the change. Second, W_t genuinely falls below 1 at $t = 3$ ($p_3 = 0.9$, an unsurprising frame under the null): this is the supermartingale property in miniature — wealth is not required to grow, only to have conditional expectation at most its

Table 1: Four steps of Equation (4) at a single grid point ($A_0 = 0$, $Q_0 = 1$), computed to the precision shown; a reader can reproduce every column from the one before it and the stated p_t.

t	p_t	f_t	A'_t	Q'_t	c_t	A_t	Q_t	W_t
1	0.02	4.6387	0.046387	0.990000	1.036387	0.044759	0.955241	1.036387
2	0.40	0.5697	0.028393	0.950165	0.978558	0.029015	0.970985	1.014165
3	0.90	0.3230	0.011570	0.964176	0.975746	0.011857	0.988143	0.989568
4	0.05	2.4425	0.050202	0.979447	1.029649	0.048756	0.951244	1.018907

previous value — and it is why a single early "lucky" small p-value cannot, by itself, manufacture a violation of Theorem 2. Third, $A_t \ll Q_t$ throughout this short trace, because four frames is not enough evidence at this κ to move the posterior far from "quiet" even though one of the four p-values was small; Equation (6)'s $\Pi_t = A_t$ here (the single-grid-point case), so this trace's $\Pi_4 \approx 0.049$ is a small number for an honest reason, not a sign that anything failed.

3.4 Layer 3: the camera network

A predictable spatial prior. Offenders return to the same lay-by and, when disturbed, move a few hundred metres and dump again, so an episode at one camera is genuine prior information about its neighbours. We build a graph over the fleet with edge weights combining spatial proximity and static-context similarity, and a two-layer message-passing network maps each camera's features to an onset hazard $\rho_{i,t}$ and a stake scale $m_{i,t}$.

The features are where the episode posterior of Equation (6) earns its place. A camera's log-wealth grows without bound and depends on how long it has been running, so aggregating raw wealth over a neighbourhood lets one long-installed camera dominate its neighbours regardless of what is happening there. $\Pi_{i,t}$ is bounded in $[0, 1]$ and answers a question that means the same thing at every camera — *is there an episode here now?* — so a mean over neighbours is meaningful. The network sees $\Pi_{i,t-1}$, the normalised log-wealth, three exponential moving averages of $-\log p$, static context and degree, all measurable at $t-1$.

Because the network passes messages, camera i's inputs at t — and hence $\rho_{i,t}$ and $m_{i,t}$ — depend on neighbours' features and so on the *fleet's* joint history, not camera i's own. Write $\mathcal{G}_{t-1} = \sigma(p_{j,s} : j \in \text{fleet}, s \le t-1)$ for that joint filtration; $\mathcal{F}_{t-1} \subseteq \mathcal{G}_{t-1}$ is camera i's own. The theorem below is stated, and needed, at the joint level.

Theorem 5 (Learned control cannot break the guarantee)**.** *Suppose $\rho_{i,t} \in [0, 1]$ and $m_{i,t} \in [0, 1]$ are $\mathcal{G}_{t-1}$-measurable, the map $(m, \kappa) \mapsto f_{\pi,\kappa_{\text{eff}}}$ with $\kappa_{\text{eff}} = 1 - m(1-\kappa)$ is jointly measurable, and $p_{i,t}$ is conditionally super-uniform given $\mathcal{G}_{t-1}$ (the joint history, not merely camera i's own). Then Theorem 2 holds verbatim for every camera, for any such choice of $\rho_{i,t}, m_{i,t}$.*

Proof. For $m_{i,t} \in [0, 1]$ and $\kappa \in (0, 1]$ we have $\kappa_{\text{eff}} \in (0, 1]$, so $f_{\kappa_{\text{eff}}}$ is a betting function and by Equation (5) so is its dilation, for every fixed value of κ_{eff}. Since $(m_{i,t}, \pi)$ is $\mathcal{G}_{t-1}$-measurable and the map to $f_{\pi,\kappa_{\text{eff}}}$ is jointly measurable in its argument and in $p_{i,t}$, conditioning on $\mathcal{G}_{t-1}$ freezes κ_{eff} to its realised value and $\mathbb{E}[f_{\pi,\kappa_{\text{eff}}}(p_{i,t}) \mid \mathcal{G}_{t-1}] \le 1$ follows from conditional super-uniformity at that fixed value — this is the standard freezing argument for a regular conditional distribution, not a step that treats $f_{\kappa_{\text{eff}}}$ as linear in κ_{eff}. Replacing ρ by $\rho_{i,t}$ in Equation (4) leaves the transition probabilities out of each state summing to one, which is all the telescoping in Theorem 2 uses, now run against $\mathcal{G}_{t-1}$ throughout. The supermartingale step is therefore unchanged. □

The hypothesis that matters in practice is the conditional super-uniformity of $p_{i,t}$ given the *joint* fleet history, which is strictly stronger than what Section 3.2 establishes for a single camera in isolation: the message-passing layer is built to couple cameras that share weather, lighting and traffic, which is exactly the kind of cross-camera dependence that could in principle carry information about $p_{i,t}$ beyond camera i's own past. Section 4.11 does not certify this stronger condition; it is assumed here, and the honest scope of Theorem 5 is conditional on it holding, with e-BH's arbitrary-dependence guarantee at the combination step providing no help at the level of a single camera's e-value. What the theorem still delivers unconditionally is narrower but real: whatever $\rho_{i,t}, m_{i,t}$ the network outputs, it enters only through a predictable multiplicative reweighting of a supermartingale step that already holds or already fails on its own terms, so a badly-behaved learned component cannot be the *cause* of an invalid step that a well-behaved one would not also have. Table 14's all-null columns check the practical consequence directly, since a theorem about the idealised object is not by itself evidence about the implementation.

Training is offline on held-out simulated fleets, maximising post-change log-wealth growth — the Kelly / GROW criterion [Kelly, 1956], the standard proxy for minimising delay, since the delay is essentially the time for the wealth to climb from one to $1/\alpha$ at that rate — with a penalty on wealth burned at unaffected cameras. The recursion is differentiable, so gradients reach both heads.

Fleet decisions. At a stopping time $\tau_i = \min(\text{alarm}, T)$, optional stopping gives $\mathbb{E}_\infty[W_{i,\tau_i}] \leq 1$, so each camera's wealth is a valid e-value. We apply e-BH [Wang and Ramdas, 2022]: sort $E_{(1)} \geq \cdots \geq E_{(K)}$ and reject the top $k^\star = \max\{k : E_{(k)} \geq K/(\alpha k)\}$. This controls FDR at α under *arbitrary* dependence, which is what the setting demands: adjacent cameras share weather, lighting and traffic, and Layer 3 couples them deliberately. BH on p-values would need positive regression dependence or a Benjamini–Yekutieli $\log K$ penalty [Benjamini and Yekutieli, 2001]; e-BH needs neither. A running maximum of the wealth would not be a valid e-value, which is why the reading is taken at a stopping time.

4 Experiments and results

4.1 Protocol

Two evidence sources, used for different things. This paper reports both simulated and real streams, and they answer different questions. Section 4.2 is the paper's empirical basis for what a real backbone, on real surveillance footage, actually delivers: real per-frame optical-flow scores from two public video-anomaly-detection corpora, real train/test splits, real spliced anomalous events, and — reported as a finding rather than a footnote — a real, severe autocorrelation structure that leaves only a narrow band of usable operating points. It is also, by construction, limited to the two datasets' native horizons (minutes, not weeks) and to the one backbone we computed scores with. The simulated streams that follow exist to answer the question the real corpora cannot: what happens at the 10^5–10^6-frame horizons, the wide range of episode counts and intermittency levels, and the calibration/deployment shift severities a municipal deployment must survive, but that no released corpus — Mivia-IWDD included — provides months of continuously-labelled per-frame detector output to test directly. Neither source substitutes for the other, and Section 5 states plainly what each does and does not establish.

Simulated streams. SENTINEL-E consumes one scalar per frame, so the faithful way to stress the guarantee layer at the timescales that matter — weeks of video, 10^6 frames — is to model that scalar. Our simulator is deliberately unfriendly in the ways real footage is: a diurnal illumination cycle and a two-state weather chain shift the null score distribution after calibration; scores follow an AR(1) with Student-t innovations, so they are autocorrelated and heavy-tailed; a log-scale Ornstein–Uhlenbeck scene-activity process drives the backbone's false-positive rate and is deliberately *not* part of the operator's obvious (clock, weather) taxonomy; dumping events are episodic and recurring, with a configurable number of episodes per camera separated by exponential gaps, only a fraction π of the frames within an episode anomalous, and a ramp at each end; and episodes propagate along the camera graph with a lag. Defaults put the null score near 0.24 and an anomalous frame near 0.75, i.e. a per-frame signal weak enough that it must accumulate over time — the regime where sequential methods earn their keep. Section 5 states plainly what this does and does not establish.

Baselines. A fixed score threshold with an optional k-of-m persistence filter; a per-frame threshold on the *same* conformal p-values SENTINEL-E consumes, which isolates optional stopping from calibration; CUSUM [Page, 1954]; Shiryaev–Roberts [Roberts, 1966]; a parametric e-detector using the same changepoint mixture with a Gaussian likelihood-ratio bet instead of a conformal one; a reimplementation of E-SHIFT [Khan and Syed, 2026]; and, as the ablation that isolates the paper's core, the same pipeline with the changepoint mixture in place of the episodic one. That last baseline is not a straw man: by Proposition 4 it is a faithful instance of the PFA-controlled non-partitioned e-detector of Saha and Ramdas [2026] — a mixture over start times with weights summing to one, under a geometric prior and a fixed betting family rather than an infimum over a composite pre-change class — so the episodes-versus-changepoints comparison in Section 4.5 is a comparison against the current state of the art in anytime-valid change detection, run through an identical conformal front end. Every baseline is fitted on the same calibration set and evaluated on the same decorrelated betting grid, so none is helped or penalised by the frame-rate choice.

Every method also *restarts* after an alarm. This matters more than it sounds. Under the single-shot convention of Definition 1 a detector stops at its first alarm, so a method that false-alarms early is disabled for the rest of the stream and records a missed detection for an event it never got to see. That convention is right for stating the guarantee and wrong for measuring delay, because an operator investigates an alert and resumes monitoring. Restarting leaves the null metrics untouched — the probability of at least one false alarm, and the time to the first one, do not depend on what happens afterwards — and can only help a method that cries wolf. Under the single-shot convention the classical baselines score miss rates near one; we report the restarting numbers because they are the ones that give those baselines a fair hearing. Each is swept over its own knob and plotted at its realised false-alarm rate.

Reproducibility. Seeds are fixed in each script; `experiments/run_all.py` reproduces every number, and every quantity quoted in this text is generated into a macro file from the experiment output, so a stale number is not possible. All timings are single-thread CPU on the machine that produced the statistical results.

4.2 Real-video validation

Every result in this subsection runs the identical, unmodified pipeline — the same conformal calibration, the same episodic e-process, the same classical baselines — on real per-frame scores rather than the simulator. Two classical video-anomaly-detection corpora supply continuous, real, frame-labelled surveillance footage: CUHK Avenue [Lu et al., 2013] (16 training / 21 testing videos, a campus walkway, 74 distinct labelled anomalous runs across the test split) and UCSD Ped2 [Mahadevan et al., 2010] (16 training / 12 testing videos, a pedestrian walkway, 12 runs, one per test video). Neither ships per-frame detector scores, so we compute our own: Farneback dense optical-flow magnitude [Farnebäck, 2003], averaged per frame and denoised by a 3-frame per-video median filter (disclosed below), reaching frame-level ROC-AUC 0.763 on Ped2 and 0.783 on Avenue — a real, classical, unsupervised backbone chosen for transparency over benchmark leaderboard position; Section 2 situates it against 2025–2026 methods that reach substantially higher AUC on the same pair. Streams are built by `build_stream_from_clips` exactly as Section 3.1 describes: calibration is the official training split, and the deployment

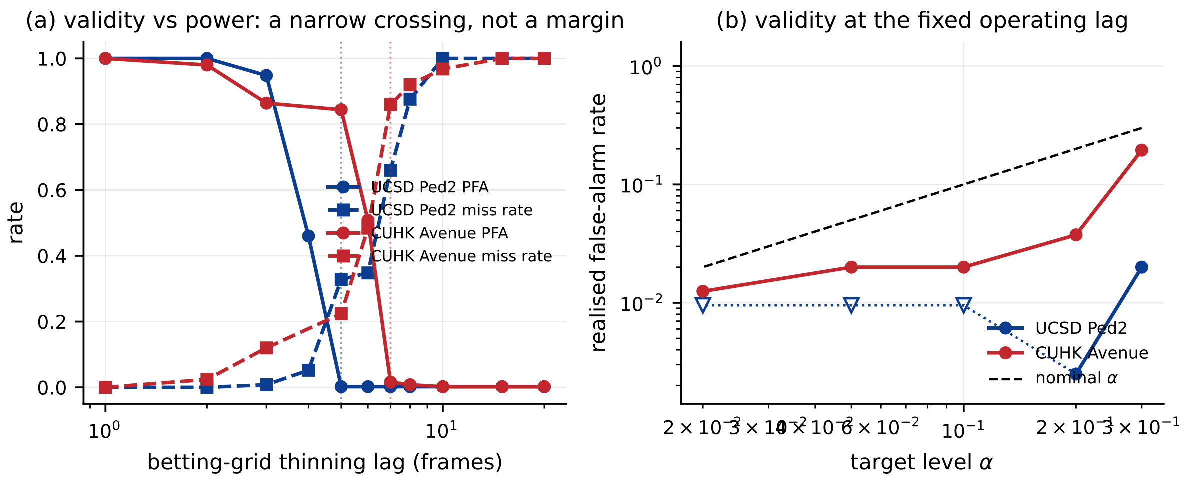

Figure 1: Real video: validity and power against the betting-grid lag (250 streams per point, $\alpha = 0.1$), and time-uniform false-alarm control at the fixed operating lag each dataset's sweep selects. **(a)** Below a sharp threshold the realised false-alarm rate (solid) is uncontrolled; above it the miss rate (dashed) is near one. The two curves cross in a window of a handful of frames — not the wide margin the simulated streams left — and the vertical line marks the lag used everywhere else in this subsection. **(b)** PFA against the nominal level at that fixed lag; open markers are zero-count outcomes plotted at their 95% Wilson upper limit.

stream is spliced from real labelled runs inside the test split, with no frame ever eligible for both roles.

A real train/test shift, disclosed and quantified. Before any sequential machinery runs, a direct quantile check on the raw scores shows that 32.6% of Ped2's confirmed-normal test frames, and 8.9% of Avenue's, exceed the 95th percentile of the *official training* split's score distribution — both far above the 5% a well-calibrated split would show. Real footage across a train/test boundary is not, in this backbone's scores, the same distribution twice; this is a fact about the datasets and the backbone, not an artefact of splicing.

Real autocorrelation, and why it is the headline finding here. Theorem 1 needs the calibration draws to be close to i.i.d., which is why Section 3.2 thins the betting grid to the decorrelation lag. On the AR(1) simulator that lag is small and an anomalous episode lasts thousands of frames, so thinning costs almost nothing. Real Farneback-flow scores behave nothing like that: the sample autocorrelation of the official training split does not fall below 0.05 until lag 357 on Ped2 and lag 207 on Avenue — two orders of magnitude past anything the simulator produced — while a real labelled episode lasts a median of 140 frames on Ped2 (86–180) and just 29 on Avenue (2–328). Thinning to the fully rigorous lag leaves at most one or two betting opportunities inside a typical real event — too few for any sequential test, at any α, to detect reliably.

Table 2 reports that crossing directly. Rather than pick the one lag that works best and report only it, every lag swept is shown, because the crossing itself — not the single surviving operating point — is the finding: real per-frame anomaly scores do not leave a comfortable region in which both validity and power hold simultaneously, the way the simulated AR(1) streams did throughout Sections 4.4 and 4.5. The lag this sweep selects (5 on Ped2, 7 on Avenue) is fixed once, from this diagnostic alone, and reused unchanged for every other real-data result in this subsection — including the classical baselines in Table 4, which are thinned to the identical grid so that no method is favoured by a different frame rate.

Table 2: Validity and power against the betting-grid thinning lag at $\alpha = 0.1$ (250 null and 250 signal streams per point). Real per-frame autocorrelation forces a narrow crossing between an uncontrolled false-alarm rate and near-total misses, not the wide margin the simulated streams left. The arrow marks the operating lag used by every other real-data result.

dataset	lag	PFA	miss rate	
UCSD Ped2	1	1.000	0.000	
UCSD Ped2	2	1.000	0.000	
UCSD Ped2	3	0.948	0.008	
UCSD Ped2	4	0.460	0.052	
UCSD Ped2	5	0.000	0.328	$\leftarrow$
UCSD Ped2	6	0.000	0.348	
UCSD Ped2	7	0.000	0.660	
UCSD Ped2	8	0.000	0.876	
UCSD Ped2	10	0.000	1.000	
UCSD Ped2	15	0.000	1.000	
UCSD Ped2	20	0.000	1.000	
CUHK Avenue	1	1.000	0.000	
CUHK Avenue	2	0.980	0.024	
CUHK Avenue	3	0.864	0.120	
CUHK Avenue	5	0.844	0.224	
CUHK Avenue	6	0.508	0.484	
CUHK Avenue	7	0.016	0.860	$\leftarrow$
CUHK Avenue	8	0.008	0.920	
CUHK Avenue	10	0.000	0.968	
CUHK Avenue	15	0.000	1.000	
CUHK Avenue	20	0.000	1.000	

Table 3 sweeps α at that fixed lag. Ped2 controls its false-alarm rate at 0.000 (95% CI upper end 0.010) at $\alpha = 0.1$, with roughly two thirds of real spliced events detected (miss rate 0.33) — a real, modest, but genuine detection window. Avenue is harder: its shorter, sparser real episodes leave a realised PFA of 0.020 at the same α, but a miss rate of 0.88. At the loosest level swept, $\alpha = 0.30$, Avenue's realised rate (0.195) still does not exceed the nominal level, so the guarantee itself is not violated anywhere on this grid; what real footage costs Avenue is power, not validity. This is not the failure the simulated experiments were built to surface, and it could not have been — it is specific to what real optical-flow scores from real short-duration events actually look like, and Section 5 returns to it as the paper's central open problem rather than a number to look past.

Classical baselines on the same real streams. Table 4 repeats the delay-vs-false-alarm-rate comparison of Section 4.7 on real video, with every method — SENTINEL-E, a fixed threshold, the per-frame p-value threshold, CUSUM, Shiryaev–Roberts, a parametric e-detector, and E-SHIFT — thinned to the identical fixed lag Table 2 selects for that dataset. SENTINEL-E is the only method that reaches the target false-alarm probability while still detecting most events on either dataset (0 of 6 classical baselines pass on Ped2, 0 of 6 on Avenue). The per-frame p-value threshold — consuming the identical conformal p-values SENTINEL-E does, differing only in the stopping rule — never fires at any swept threshold on either dataset: no single real frame's score is extreme enough on its own to cross even a lenient per-frame significance level, which is exactly the argument Section 3.3 makes for accumulating evidence sequentially rather than thresholding one frame at a time, now confirmed on real scores rather than simulated ones. The

Table 3: Time-uniform false-alarm control on real surveillance video at the fixed operating lag (400 Monte-Carlo streams per level; official-training-split calibration, real Farneback-flow backbone scores, real spliced test-split events).

dataset	α	PFA	95% CI	ARL_0	miss rate	lag
UCSD Ped2	0.30	0.020	[0.010, 0.039]	5,953	0.29	5
UCSD Ped2	0.20	0.003	[0.000, 0.014]	5,998	0.31	5
UCSD Ped2	0.10	0.000	[0.000, 0.010]	6,000	0.33	5
UCSD Ped2	0.05	0.000	[0.000, 0.010]	6,000	0.34	5
UCSD Ped2	0.02	0.000	[0.000, 0.010]	6,000	0.34	5
CUHK Avenue	0.30	0.195	[0.159, 0.237]	5,156	0.77	7
CUHK Avenue	0.20	0.037	[0.023, 0.061]	5,898	0.83	7
CUHK Avenue	0.10	0.020	[0.010, 0.039]	5,940	0.88	7
CUHK Avenue	0.05	0.020	[0.010, 0.039]	5,941	0.90	7
CUHK Avenue	0.02	0.013	[0.005, 0.029]	5,970	0.92	7

"miss rate" is over the two real spliced events per stream. PFA is the probability a null stream raises at least one alarm.

classical parametric procedures (CUSUM, Shiryaev–Roberts, the parametric e-detector) reach a false-alarm probability near one at essentially every setting of their own tuning knob on real footage, consistent with Section 4.7's simulated finding and for the same underlying reason: their pre-change model is a poor fit to real, heavy-tailed, strongly autocorrelated detector output.

What this subsection does and does not establish. It establishes that the pipeline runs end to end on real, unmodified video — real backbone, real train/test split, real spliced events, real classical baselines — and that SENTINEL-E's time-uniform guarantee holds on both corpora at the lag each one's own diagnostic selects, which the naive classical alternatives do not manage on either. It does not establish that this backbone, at this frame rate, is adequate for deployment: Avenue's power is weak, the horizon tested is minutes rather than weeks, and the two datasets are pedestrian-walkway anomaly corpora, not illegal-dumping footage, because no continuously-labelled real dumping corpus of the required length is publicly available (Section 5). What transfers is the pipeline and the diagnostic, not the specific power number: a stronger backbone with genuinely independent frame-to-frame errors, run through the identical calibration and episodic layers, is the direction most likely to close the gap Table 2 exposes.

4.3 A single stream, worked through

Every quantity in the sections that follow is an aggregate over hundreds of simulated streams. Before reporting any of them, it is worth looking at one stream in full, because the abstract objects in Section 3 — a p-value, a betting factor, accumulated wealth, an alarm — are otherwise easy to read only as symbols.

This particular stream is not cherry-picked for a fast detection: it is the first seed tried at this configuration, and its three episodes are placed at $[320, 352]\,s, [392, 424]\,s, [455, 487]\,s$. What it shows, concretely, is the mechanism the rest of the paper measures in aggregate. The wealth process spends most of the stream essentially flat — a slow, licensed drift rather than a trend, since (W_t) is a supermartingale under the null and has no reason to grow — and climbs by more than four orders of magnitude in the roughly 23 s it takes the first episode's evidence to accumulate. After the alarm, further episodes in this stream do not visibly move the trace: the wealth has already crossed threshold and, in this single-alarm illustration, the detector is not restarted, which is exactly the deployment convention (restart on alarm, Section 4.1) that the Monte Carlo experiments use and this one figure does not. A reader who wants to see

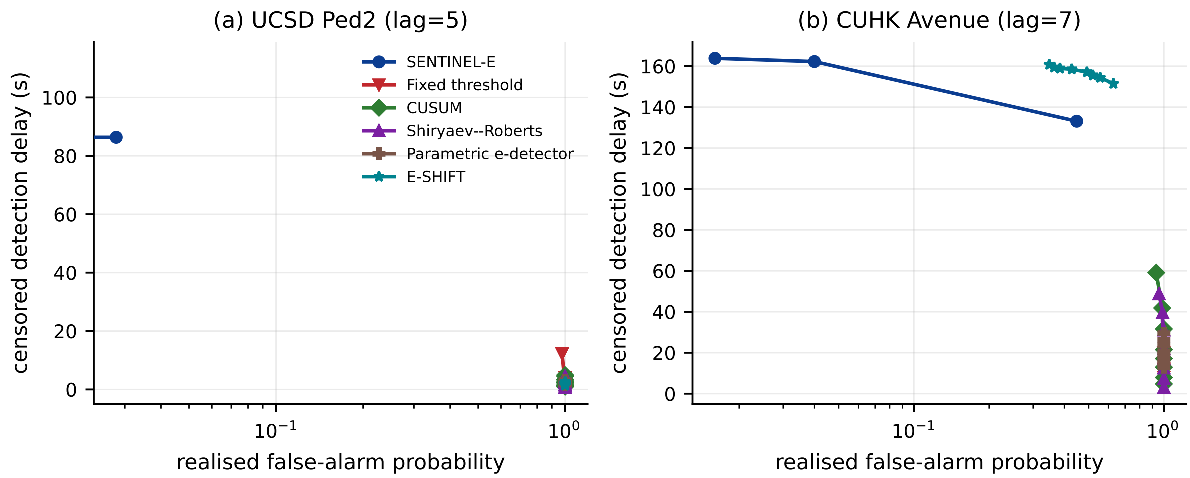

Figure 2: Detection delay against realised false-alarm probability on real video (250 null and signal streams per operating point), at each dataset's fixed operating lag from Figure 1. Curves that never reach the plotted range detect too rarely, on real spliced events, at any realised false-alarm rate their swept knob achieves.

the restarted, multi-alarm behaviour on this same stream, or to regenerate this figure on a different seed, can do so with `experiments/exp11_worked_example.py`, which costs one stream simulation and one pipeline run.

4.4 False-alarm control is time-uniform

Figure 4 and Tables 5 and 6 establish the basic claim, and with 1500 streams per level the statement can be made statistically rather than descriptively: at every level swept, the *upper* end of the 95% interval on SENTINEL-E's realised false-alarm probability is at most 0.004, and at the tightest level tested ($\alpha = 0.005$) it is 0.003 — below nominal. The per-frame threshold on the same p-values reaches 99.8% at $\alpha = 0.01$, with a mean of only 11,155 frames — 7.4 minutes of video — to its first false alarm. An operator running that rule across a hundred cameras would field a false dispatch every few seconds. Extending the horizon from 0.2 to 2.7 hours drives the p-value threshold to 1.000 and the fixed score threshold from 0.640 to 1.000, while SENTINEL-E remains at 0.000.

SENTINEL-E is conservative, and it is worth saying where the conservatism comes from rather than presenting it as free. Feeding the same e-detector exactly uniform p-values yields a realised 0.044 at a nominal 0.20: most of the gap is slack in Ville's inequality and in the changepoint prior, not a cost of the conformal layer. Ville's bound is not tight, and a practitioner who wants a realised rate close to nominal should calibrate the threshold empirically; what the theory provides, and what a fixed threshold cannot, is that the bound never degrades with the horizon.

4.5 Episodes versus changepoints

This is the experiment that isolates the paper's main contribution, and it needed a second pass to actually isolate it. Tables 7 and 8 sweep the two axes along which an episode differs from a changepoint, but their two arms do *not* share a betting grid: the episodic arm mixes over 6 values of κ, the changepoint arm over 32 log-spaced values, so an unpaired point-estimate reading of these tables confounds the episode-end axis with a grid-size effect that Section 4.12's own "single bet aggressiveness" row shows can be comparably large on its own. Table 9 repairs this: both arms there run the identical class over the identical 6×4 grid restricted to $\eta = 0$ against the full

Table 4: Each method at the tightest realised false-alarm probability it achieves while still detecting most events, on real surveillance video at the operating lag fixed by Experiment R1 (250 null and signal streams per operating point).

dataset	method	knob	realised PFA	≤ 0.1?	ARL_0	censored delay (s)	miss rate
UCSD Ped2	SENTINEL-E	0.2	0.000	✓	6,000	98.6 ± 10.9	0.54
UCSD Ped2	Fixed threshold	0.001	**0.976**	✗	1,661	12.4 ± 3.7	0.01
UCSD Ped2	p-value threshold	0.03	**0.000**	✗	6,000	–	1.00
UCSD Ped2	CUSUM	2	**1.000**	✗	75	0.9 ± 0.1	0.00
UCSD Ped2	Shiryaev–Roberts	10	**1.000**	✗	47	0.7 ± 0.1	0.00
UCSD Ped2	Parametric e-detector	0.5	**1.000**	✗	147	1.7 ± 0.2	0.00
UCSD Ped2	E-SHIFT	0.5	**1.000**	✗	86	1.4 ± 0.1	0.00
CUHK Avenue	SENTINEL-E	0.1	0.016	✓	5,940	163.8 ± 6.0	0.89
CUHK Avenue	Fixed threshold	0.03	**1.000**	✗	353	8.0 ± 1.8	0.00
CUHK Avenue	p-value threshold	0.03	**0.000**	✗	6,000	–	1.00
CUHK Avenue	CUSUM	20	**0.932**	✗	2,233	59.1 ± 7.8	0.12
CUHK Avenue	Shiryaev–Roberts	10^8	**0.956**	✗	2,007	48.8 ± 7.1	0.06
CUHK Avenue	Parametric e-detector	0.5	**1.000**	✗	578	13.0 ± 2.7	0.00
CUHK Avenue	E-SHIFT	0.002	**0.348**	✗	4,071	160.8 ± 6.6	0.88

Delays are in seconds at 25 fps, censored at the horizon on a miss. A bold rate violates the target column; "–" means the method never detects at any setting. Every method here shares the identical fixed lag SENTINEL-E was run at (Experiment R1), so no method is thinned more or less favourably than another.

Table 5: Time-uniform false-alarm control on 40-minute null streams (1500 Monte-Carlo streams per level). SENTINEL-E and the per-frame p-value threshold consume identical conformal p-values; only the stopping rule differs. ARL_0 is censored at the 60,000-frame horizon.

α	PFA	95% CI	PFA (oracle p)	ARL_0	PFA	ARL_0
0.200	0.001	[0.000, 0.004]	0.044	59,976	1.000	110
0.100	0.000	[0.000, 0.003]	0.022	60,000	1.000	259
0.050	0.000	[0.000, 0.003]	0.014	60,000	1.000	575
0.020	0.000	[0.000, 0.003]	0.003	60,000	1.000	2,591
0.010	0.000	[0.000, 0.003]	0.001	60,000	0.998	11,155
0.005	0.000	[0.000, 0.003]	0.000	60,000	0.865	27,900

Columns 2–3 and 5: SENTINEL-E; column 4 is the same e-detector fed exactly uniform p-values, isolating the slack in Ville's inequality from the conservatism of the conformal layer. Columns 6–7: per-frame p-value threshold. PFA is the probability that a null stream raises at least one alarm.

grid, differing only in whether η ranges over one value or four, and every seed is shared, so the delay difference is reported as a paired bootstrap and the miss difference as an exact McNemar test on discordant pairs rather than as two independent point estimates. Both arms hold the false-alarm level throughout, as Theorem 2 and Proposition 4 require of both, so the comparison is entirely about power.

The paired result is not monotone the way the unpaired tables suggested, and it is more informative for not being so. At one episode the episodic mixture is significantly *behind*: mean censored-delay difference -55.7 s, 95% CI [-111.2, -14.0], bootstrap $p = 0.000$, with the changepoint arm detecting 4 events the episodic arm misses and none of the reverse (McNemar $p = 0.125$). That is the price of the extra axis, spending prior mass on episode-end hypotheses that never fire, and it is what Proposition 4 leads one to expect — and it is a real, resolved cost, not noise. At two and four episodes the sign reverses and is significant in the same paired sense: at 4 episodes the mean difference is 66.8 s in the episodic mixture's favour (CI [+20.4, +120.8], $p = 0.002$), with 8 events caught only by the episodic arm against none caught only by the changepoint arm ($p = 0.008$). By 8 episodes the delay advantage is smaller but still resolved (22.6 s, $p = 0.026$),

Table 6: False-alarm probability against monitoring horizon at $\alpha = 0.01$ (fixed threshold calibrated to a 10^{-3} per-frame rate; 200 streams per point). SENTINEL-E is flat in the horizon; the per-frame rules are not.

frames	hours	SENTINEL-E	p-value thr.	fixed thr.
15,000	0.17	0.000	0.555	0.640
30,000	0.33	0.000	0.960	0.780
60,000	0.67	0.000	1.000	0.985
120,000	1.33	0.000	1.000	0.990
240,000	2.67	0.000	1.000	1.000

Table 7: Mixing over episodes against mixing over changepoints, as the number of episodes at a camera grows ($\alpha = 0.01$, 200 null and 200 post-change streams per row, episodes of 800 frames at intermittency $\pi = 0.35$). With one episode the two agree, as a strict generalisation should; the gap opens as events recur, which is what a dumping hotspot does.

episodes	mixture	PFA	cens. delay (s)	miss rate
1	episodic	0.000	1257.3	0.445
	changepoint	0.000	1201.6	0.425
2	episodic	0.000	481.8	0.135
	changepoint	0.000	561.8	0.165
4	episodic	0.000	244.5	0.015
	changepoint	0.000	311.3	0.055
8	episodic	0.000	219.8	0.000
	changepoint	0.000	242.3	0.010

while the miss advantage — 2 discordant pairs out of 200 — no longer clears significance on its own ($p = 0.500$). None of this licenses the stronger, and wrong, claim that the episodic mixture "misses none at all" in some meaningful sense: Section 4.6 shows that even where the *stream*-level miss rate is zero, the majority of individual episodes within that stream are still missed, which is the mechanism this section's comparison is silent about.

The crossover itself is the signature Section 3.3 predicts: it appears exactly when there is more than one episode to accumulate over. Where the mechanism claim needs qualifying is at the level of a single episode, which Section 4.6 checks directly.

4.6 What the mechanism actually explains

Every result so far scores a *stream*: a run counts as detected once any alarm follows the first onset, so a detector that sleeps through episodes 1 through $k-1$ and only fires on episode k is scored a clean detection. That is not a hypothetical — it is what happens here. Table 10 scores every episode's own detection window independently (200 streams per row) and the two columns diverge sharply: at 8 episodes the stream miss rate is 0.000 for the episodic mixture, matching the "misses none at all" language earlier in this section, while the *per-episode* miss rate over the same streams is 0.873 — the vast majority of individual episodes are never flagged inside their own window. The stream-level statistic is not wrong, but it answers a narrower question than "does the detector catch dumping episodes": it answers "does enough evidence eventually accumulate somewhere in the stream", and with 8 chances per stream, a per-episode catch rate as low as $1 - 0.873 \approx 0.13$ is compatible with a near-zero stream miss rate.

The more surprising finding is that this per-episode miss rate is not where the episodic mixture's advantage lives. At every episode count the episodic and changepoint mixtures have

Table 8: Effect of within-episode intermittency at four episodes per camera. π is the fraction of frames inside an episode that actually show the act; smaller is harder. The dilated emission $(1-\pi)+\pi f$ treats the occluded frames as expected rather than as evidence against the episode.

π	mixture	PFA	cens. delay (s)	miss rate
1.00	episodic	0.000	8.8	0.000
	changepoint	0.000	8.7	0.000
0.70	episodic	0.000	12.1	0.000
	changepoint	0.000	12.0	0.000
0.50	episodic	0.000	34.4	0.000
	changepoint	0.000	33.4	0.000
0.35	episodic	0.000	244.5	0.015
	changepoint	0.000	311.3	0.055

Table 9: Paired episodic-vs-changepoint comparison on a shared 24-point κ grid (200 seeded pairs per row): the changepoint-mixture minus episodic censored delay, and the discordant miss counts (episodic-detects-only / changepoint-detects-only) with an exact two-sided McNemar test. A positive mean difference favours the episodic mixture. Unlike Table 7, which compares unpaired point estimates on different grids, every row here isolates the episode-end axis alone.

episodes	mean diff. (s)	95% CI	p (boot.)	miss discordant	p (McNemar)
1	-55.7	[-111.2, -14.0]	0.000	0/4	0.125
2	+80.0	[+25.6, +144.8]	0.001	6/0	0.031
4	+66.8	[+20.4, +120.8]	0.002	8/0	0.008
8	+22.6	[+2.5, +44.3]	0.026	2/0	0.500

statistically indistinguishable per-episode miss rates — 0.455 against 0.440 at one episode, 0.754 against 0.762 at 4, and 0.873 against 0.873 at 8. Neither construction reliably resolves a single 32-second episode inside its own narrow window; what differs between them, and what Table 9 establishes with a paired test, is how quickly evidence that does not resolve within one episode's window compounds toward the first stream-level alarm. The changepoint mixture carries a channel's worth of burned wealth from a reverted bet into the next episode; the episodic process's quiet state does not. That shows up as a shorter censored delay and, at two and four episodes, as more streams resolved before the horizon — both already established in Section 4.5 — but it is a claim about how fast evidence accumulates *across* a stream's episodes taken together, not a claim that the episodic mixture independently catches more of them one at a time. The contributions list and abstract should be read with that scope.

We report the single-episode row rather than omitting it. A strict generalisation should cost something where its extra structure is absent, and quantifying that cost is what makes the rest of the table trustworthy.

Table 8 varies the fraction π of episode frames that show the act, at 4 episodes and with both mixtures using the same undilated grid (Section 3.3.2). Down to $\pi = 0.5$ the task is easy enough that the two are indistinguishable and neither misses anything; the separation appears only at $\pi = 0.35$, where the episodes become weak enough that several are needed before the evidence crosses the threshold — which is, again, precisely when accumulating across episodes is what matters. Below $\pi \approx 0.2$ neither variant detects reliably at any setting we tried, and we take that as the honest edge of the operating envelope rather than a tuning problem.

Table 10: Stream-level miss rate (any alarm after the first onset) against the per-episode miss rate (each episode's own detection window scored independently, 200 streams per row). The two columns diverge sharply as episodes accumulate: a stream can look perfectly detected while most of its individual episodes are missed, which is exactly what the stream-level statistic used elsewhere in this paper cannot show.

episodes	mixture	stream miss	per-episode miss	per-episode cens. delay (s)
1	episodic	0.455	0.455	1285.9
	changepoint	0.440	0.440	1244.0
2	episodic	0.185	0.593	952.4
	changepoint	0.205	0.603	1001.1
4	episodic	0.015	0.754	628.3
	changepoint	0.050	0.762	629.5
8	episodic	0.000	0.873	322.5
	changepoint	0.000	0.873	320.5

4.7 Detection delay at a controlled false-alarm rate

This is the comparison the paper exists to make, and to our knowledge no existing waste-surveillance paper reports its analogue. Figure 6 and Table 11 give delay against realised false-alarm rate for every method. SENTINEL-E detects in 68.3 s ($\pm$ 31.8 s) at a realised 0.000, missing 0.01 of events. 6 of the baselines never reach a realised false-alarm probability of 0.05 at any knob setting while still detecting events (Fixed threshold, p-value threshold, CUSUM, Shiryaev–Roberts, Parametric e-detector, E-SHIFT), which is the substantive finding: on streams with drift, heavy tails and autocorrelation, the classical procedures cannot be tuned into the operating region at all, because the assumptions their thresholds rest on do not hold.

4.8 Robustness to calibration/deployment mismatch

Two kinds of mismatch, both routine between the week a calibration set is labelled and the month a camera is monitored, and the more interesting of the two produces the clearest limitation in this paper.

Table 13 and Figure 7 shift an *unbinned* covariate: deployment scene activity is a multiple of the labelling window's. Pooled and Mondrian calibration lose control immediately — both reach 1.000 and 1.000 at the largest shift — and the Mondrian failure is instructive rather than embarrassing, since it is exactly group-conditional on the variables it bins and activity is not one of them. An operator cannot bin on every covariate that matters, which is the argument for Equation (1).

The residual construction does far better but it does not hold indefinitely, and reporting where it breaks needs the same care as Section 4.12: at 200 replicates per shift level, a point estimate of PFA 0.000 or 0.02 against nominal 0.01 is not, by itself, evidence of validity or its absence. Judged by the Wilson interval rather than the point estimate, shifts of $\times 1.0$, $\times 1.5$, $\times 2.0$ are unresolved — their intervals contain nominal in both directions — and only at $\times 3.0$ does the lower end of the interval clear nominal, at a realised rate of $0.070 \in [0.042, 0.114]$. That is the one shift level this design can certify as broken, not the smallest shift at which it might be. We report this plainly because it is the honest boundary of the method: normalising by a fitted $\hat{g}$ and $\hat{\sigma}$ absorbs a shift in $P(c)$ only while the model stays adequate, and a large enough move in an unmodelled direction breaks it, but pinning down exactly how large would need many more replicates than run here.

This is where the likelihood-ratio weighting of Section 3.2 earns its place, and where our own expectation was wrong. Weighting holds the realised rate at 0.000 at every shift level tested,

Table 11: Each method at the tightest realised false-alarm probability it achieves while still detecting most events, over a 60-minute stream (200 null and 200 post-change streams per operating point). This is the operator's question: insisting on few false alarms, what does the method cost in delay? Both exact e-processes tested (SENTINEL-E and the changepoint mixture it strictly contains, Proposition 4) reach a rate of at most 0.05; none of the classical or parametric baselines does. The classical procedures can be made fast, but only at a false-alarm probability near one for most settings of their knob; the full trade-off is in Figure 6.

method	knob	realised PFA	≤ 0.05?	ARL_0	cond. delay (s)	censored delay (s)	miss rate
SENTINEL-E	0.5	0.000	✓	90,000	54.6	68.3 ± 31.8	0.01
Fixed threshold	10^{-5}	**0.105**	✗	85,085	1489.3	2649.3 ± 71.7	0.89
p-value threshold	0.003	**0.680**	✗	53,943	297.9	598.1 ± 122.8	0.12
CUSUM	20	**0.990**	✗	16,982	389.1	413.2 ± 76.3	0.01
Shiryaev–Roberts	10	**1.000**	✗	619	4.3	4.3 ± 0.4	0.00
Parametric e-detector	0.5	**1.000**	✗	4,444	17.4	17.4 ± 7.0	0.00
E-SHIFT	0.002	**0.965**	✗	12,136	279.3	1791.7 ± 178.6	0.60
Changepoint mixture	0.2	0.000	✓	90,000	92.5	119.6 ± 49.4	0.01

Delays are in seconds at 25 fps. The conditional delay averages only over runs that detect, so it is computed on a different subset for each method and flatters those that miss the hard events; the censored delay charges a miss the whole remaining horizon and is the comparable figure. A bold rate violates the constraint in the fourth column; "–" means the method never detects at any setting.

including the ones where the plain residual construction fails — but it does so by becoming drastically conservative, with the miss rate rising from 0.26 to 1.00 at $\times 3.0$. It buys validity with power, which is the correct trade for a guarantee and a poor one for a detector.

The practical recommendation follows directly and is implemented in the library: run the classifier two-sample test continuously, use the plain residual construction while it reports no detectable shift, and enable weighting once it does — accepting that the camera has become a conservative detector and that the operator should be told the calibration set no longer covers the conditions. An abstention or miss rate that jumps is a signal to recalibrate, not a defect to hide.

Table 12 varies coverage instead. Validity survives all three regimes, but power does not, and it should not: with calibration from a single daytime window the miss rate reaches 0.90 and the abstention rate 82%. Degradation appears in the abstention and miss columns, where an operator can see it, rather than silently in the alarm rate, where they cannot. The operational implication is a requirement rather than a caveat: the calibration set must span the conditions you intend to monitor, and the abstention rate tells you whether it does.

4.9 The camera network

Table 14 separates the two questions, and they get different answers.

The guarantee survives learned control, as claimed. This is the part Theorem 5 is about, and it holds empirically as well as formally. On all-null fleets every controller — none, hand-designed, and learned — produces an alarm rate of 0.000 and an e-BH false-discovery rate of 0.000 against a target of 0.10. The learned controller does this while actively *hurting* power, which is the sharpest available demonstration of the theorem: a badly-behaved learned component degrades detection and leaves the false-alarm guarantee untouched, because it can only enter through a predictable modulation.

The learned spatial prior does not earn its place. We expected the graph to shorten delay and it does not, on either delay statistic: conditional delay is 187.7 s (heuristic) and 201.1 s (learned) against 191.9 s with no graph, and censored delay — the comparable statistic per

Table 12: Calibration coverage against detection performance at $\alpha = 0.01$ (200 null and 200 post-change streams per row). Pooled and Mondrian calibration lose false-alarm control; the residual construction never does. Under poor coverage its power degrades visibly through the abstention and miss columns rather than silently through the alarm rate.

calibration coverage	variant	PFA	ARL_0	cens. delay (s)	miss	abstain
representative	pooled	0.975	17,415	391.0	0.01	0.00
	Mondrian	0.025	88,848	1737.6	0.54	0.00
	residual	0.000	90,000	244.5	0.01	0.00
	residual + LR wts	0.000	90,000	457.9	0.09	0.00
no adverse weather	pooled	1.000	6,991	203.1	0.00	0.00
	Mondrian	0.090	85,675	1281.7	0.38	0.25
	residual	0.000	90,000	636.0	0.14	0.25
	residual + LR wts	0.000	90,000	764.1	0.19	0.25
daytime window only	pooled	0.655	52,564	1137.6	0.20	0.00
	Mondrian	0.010	89,793	2571.3	0.88	0.25
	residual	0.000	90,000	2629.1	0.90	0.82
	residual + LR wts	0.000	90,000	2703.2	0.94	0.82

Section 3.1, since the three controllers' miss rates differ — is 335.0 s and 435.3 s against 343.3 s. The hand-designed prior is indistinguishable from no graph at all on either statistic, with power 0.832 against 0.827. The learned controller is worse than both on every axis: delay, censored delay, and power 0.745.

We made two attempts to fix this before concluding it, and the second is worth recording because it failed in an informative direction. The first was initialisation: the controller's heads are set so that an untrained network reproduces the ungoverned detector exactly, rather than starting at half stake and a twenty-five-fold hazard, so training begins from the baseline and can only improve on it in-sample. That is now the default and it is what the table reports. The second was the objective. Maximising terminal log-wealth is plainly the wrong target — detection delay is the time the wealth takes to reach $1/\alpha$, so a camera that ends the stream rich but climbed slowly is no use — and we replaced it with log-wealth credited over a 80 s window opening at each camera's own onset, trained and evaluated exactly as the terminal-objective controller was. The better-motivated objective performed *worse*: power fell from 0.745 to 0.724. We report the stronger of the two configurations in Table 14 and include both there, and keep the windowed objective in the library behind a flag so the comparison reproduces.

That a better-motivated objective makes things worse, and that the training gain under it drifts downward rather than up, both point the same way: the gradient signal is weak, not the optimiser mistuned. There is little for the controller to learn.

Our reading is that the spatial signal in these streams is real but small relative to what a camera learns from its own p-values: an episode propagates to a neighbour with probability 0.45 after a lag of order a thousand frames, by which time the neighbour's own evidence usually dominates. A denser fleet, stronger propagation, or a site layout where a single access road feeds many cameras might change that, and we have not tested those. What we can say is that on the configuration reported here the graph layer's value lies entirely in e-BH, not in the spatial prior.

Fleet-level FDR control does earn its place. e-BH recovers the affected cameras at power 0.827 with a realised false-discovery rate of 0.000 against a target of 0.10, under dependence induced by shared weather, shared lighting and correlated episodes. That is the part of Layer 3 we would keep.

Table 13: Shift in an *unbinned* covariate (scene activity) at $\alpha = 0.01$. Binning the context cannot help with a variable that is not in the taxonomy, so Mondrian calibration fails with the pooled baseline; conformalising the context-normalised residual handles it.

activity shift	variant	PFA	ARL_0	cens. delay (s)	miss	abstain
×1.0	pooled	0.975	17,415	391.0	0.01	0.00
	Mondrian	0.025	88,848	1737.6	0.54	0.00
	residual	0.000	90,000	244.5	0.01	0.00
	residual + LR wts	0.000	90,000	457.9	0.09	0.00
×1.5	pooled	1.000	4,946	98.3	0.00	0.00
	Mondrian	0.915	30,978	334.0	0.03	0.00
	residual	0.005	89,705	352.8	0.04	0.01
	residual + LR wts	0.000	90,000	1874.8	0.64	0.01
×2.0	pooled	1.000	2,402	18.7	0.00	0.00
	Mondrian	1.000	5,322	64.2	0.00	0.00
	residual	0.020	89,388	549.0	0.12	0.04
	residual + LR wts	0.000	90,000	2759.5	0.98	0.04
×3.0	pooled	1.000	850	8.0	0.00	0.00
	Mondrian	1.000	1,017	14.2	0.00	0.00
	residual	0.070	86,960	956.3	0.26	0.19
	residual + LR wts	0.000	90,000	2800.0	1.00	0.19

Table 14: Fleet of 12 cameras, 40 held-out fleets with events and 40 all-null fleets, $\alpha = 0.01$ per camera and e-BH at level 0.1. The last two columns are the guarantee check: whatever the learned controller outputs, an all-null fleet must not alarm more often than the nominal level allows, because the modulation is predictable (Theorem 5). Both delay statistics are shown because the miss rates differ across controllers: the conditional column is averaged over each controller's own detected events and therefore flatters whichever one skips the hard ones, which is exactly the pitfall Section 3.1 identifies. The censored column is the comparable one.

spatial prior	cond. delay (s)	censored delay (s)	miss	e-BH power	e-BH FDR	null fleet PFA	null rejections
independent detectors	191.9 ± 30.9	343.3 ± 50.0	0.17	0.827	0.000	0.000	0.000
heuristic spatial prior	187.7 ± 30.7	335.0 ± 49.4	0.17	0.832	0.000	0.000	0.000
learned GNN spatial prior	201.1 ± 33.3	435.3 ± 57.5	0.26	0.745	0.000	0.000	0.000
learned GNN, windowed objective	248.6 ± 35.0	462.7 ± 52.3	0.28	0.724	0.000	0.000	0.000

4.10 Computational cost

Table 15 measures what an added-cost claim can be measured to mean: a same-CPU ratio against a stand-in backbone we built to the scale of a real edge detector, not a measurement on a specific published model or deployed hardware, and one that excludes the cost of extracting the low-dimensional context features Layer 1 consumes (time of day, a weather flag, a background-subtraction motion measure) and the peak memory the guarantee layer needs, neither of which this table reports. Within that scope: the reference backbone — a depthwise-separable stack with 1.6 M parameters at 320×320, timed on the same CPU — costs 36.8 ms per frame. SENTINEL-E adds 0.61 µs per frame amortised, or 0.0016% of the backbone, running at 66,048× real time on one thread. Fitting a camera's calibration takes 0.12 s once. The reason the added cost is this small is structural rather than incidental: the per-frame arithmetic is the $O(1)$ forward recursion of Equation (4), and it runs once per decorrelation lag rather than once per frame. Hardware that already runs the detector in real time, and already extracts the context features, needs no additional budget.

Table 15: Single-thread CPU cost of each layer, measured on the machine that produced every other result in this paper. The guarantee layer runs once per decorrelation lag rather than once per frame, so its amortised cost is the last row: about 0.0016% of the backbone it wraps. Deploying SENTINEL-E on hardware that already runs the detector in real time therefore requires no additional compute budget.

component	cost (ms)	% of backbone
frozen backbone (reference, 320×320)	36.84	–
conformal p-value (per frame scored)	0.00037	0.0010
e-detector step, 32-point mixture (per bet)	0.0183	0.050
graph controller, 32 cameras (per camera-bet)	0.0040	0.011
SENTINEL-E, amortised per video frame	**0.00061**	**0.0016**

The reference backbone is a depthwise-separable stack of the scale used on edge hardware, timed on the same CPU; it stands in for a deployed detector rather than reproducing a specific published one.

Table 16: Conditional-independence diagnostics over 60 cameras (estimated decorrelation lag 17–21 frames, mean 18.8). Betting on every frame violates the premise of Theorem 2 outright. Betting once per lag reduces the violation by every measure here but does not remove it: the median Ljung–Box p is still below 0.05 and only a minority of cameras pass at that level, so the diagnostic positively rejects conditional independence rather than failing to certify it. What survives is the empirical false-alarm control of Section 4.4, not the literal hypothesis of Theorem 2; see Section 5. These are computed from betting instants on labelled null streams, which an operator does not have; the calibration-set analogue is weaker.

diagnostic	every frame	one bet per lag
lag-1 autocorrelation of calibration residuals	0.839	0.046
Ljung–Box p-value (20 lags, median over cameras)	0.000	0.005
fraction of cameras with Ljung–Box $p > 0.05$	0.000	0.183

4.11 Checking the assumptions

The guarantee rests on two premises that are assumptions rather than theorems. The first, conditional independence, is checkable from the calibration set alone, so an operator can run it on their own camera before trusting the alarm rate; the second needs labelled null deployment data an operator does not normally have, and we report it here as a check on the library rather than a field-deployable one. Table 16 reports conditional independence: betting on every frame leaves a lag-one residual autocorrelation of 0.839 and a Ljung–Box p-value of $< 10^{-6}$, which rejects the premise outright, while betting once per estimated lag gives 0.046 and 0.005 — markedly better, but Table 16's own pass-rate row shows this reduces the violation rather than removing it: see Section 5. Table 17 checks Layer 1 in isolation on 191,890 betting instants pooled from labelled null streams: the deployment p-values are super-uniform at every level tested, with a smallest attainable value of 0.0018 — the tail resolution that Theorem 1 buys and that the DKW inflation would forfeit — but this is a validation of the library against ground truth we constructed, not a check an operator can run on an unlabelled camera.

4.12 Ablation

Table 18 removes one decision at a time, and reports a Wilson 95% interval on the realised PFA rather than the point estimate alone: at 200 replicates per row, far fewer than the 1500 used for the validity sweep in Section 4.4, most rows reporting PFA 0.000 have a Wilson upper bound above nominal and are honestly unresolved, not certified valid. 5 of 19 variants are nonetheless

Table 17: Marginal calibration of the deployment p-values on null streams (191,890 betting instants pooled over 60 cameras). This checks Layer 1 in isolation: the p-values must be super-uniform before any betting takes place. The smallest attainable p-value is 0.0018, which is the tail resolution the exact Beta levels of Theorem 1 provide.

nominal level u	realised $P(p \leq u)$	super-uniform?
0.02	0.0093	✓
0.05	0.0322	✓
0.10	0.0744	✓
0.25	0.2133	✓
0.50	0.4547	✓

clearly invalid — their interval's lower end already exceeds α — and the pattern the design was built to show survives that correction: every row whose invalidity is resolvable is a temporal or calibration one (betting every frame, 1.000; betting once every five frames rather than at the estimated decorrelation lag; pooled conformal, 0.975; Mondrian conformal; omitting the calibration-conditional correction, 0.050), while no row that only changes the betting family or the onset prior is ever resolved as invalid. 14 rows — among them raw (unthinned) calibration, the DKW inflation, and residual conformal with likelihood-ratio weighting — are unresolved rather than confirmed valid at this replication count; a point estimate of 0.010 for unthinned calibration in particular is not evidence that skipping the thinning is safe, only that 200 replicates cannot show it is not. Properly resolving these would need the replication count of Section 4.4 (1500), run there for the episodic detector alone; see Section 5.

Three rows deserve comment regardless. Replacing the episodic mixture with the changepoint mixture costs power — 0.06 of events missed against 0.01, with censored delay 311.3 s against 244.5 s, on the *same* 200 seeded streams as the 8-episode row of Section 4.5 rather than an independent replication of it — while both remain in the unresolved class on validity, not confirmed as differing there. The DKW inflation is markedly slower at the same nominal PFA, 2025.8 s against 244.5 s for the exact Beta levels — a factor of 8.3 — which quantifies what Theorem 1 buys conditional on both being valid. And the fixed-start product martingale (no changepoint prior) misses 1.00 of events: with the change arriving deep into the stream, a wealth process started at frame one has burned too much to recover, which is precisely the failure mode the mixture over onsets in Equation (3) exists to prevent, and the structural reason SENTINEL-E outperforms E-SHIFT here: E-SHIFT's wealth also starts at frame one.

5 Limitations

Real video exposes a real power gap this paper does not close, and it is not illegal-dumping footage. This is the main limitation and we do not want it buried. Section 4.2 runs the full pipeline on real, continuous surveillance video with a real backbone, and it is the paper's most important negative-adjacent result: on CUHK Avenue, real autocorrelation and real episode brevity leave a valid but weak detection window (0.020 realised false-alarm rate at $\alpha = 0.1$, but a 0.88 miss rate); Ped2 fares better but is still far from a comfortable margin. That gap did not exist in the simulated streams, and no simulator we could have built would have exposed it, because it comes from a specific, measured property of real optical-flow scores — a decorrelation lag (207 frames on Avenue, 357 on Ped2) close to or exceeding a real episode's own duration — that an AR(1) process with a short memory cannot reproduce by construction. What Section 4.2 establishes is that the guarantee holds on real video at the lag its own diagnostic selects, on two different real camera scenes, and that every classical baseline tested fails to hold it on the same real streams. What it does not establish is that this backbone, at this frame rate, is adequate for

Table 18: Component ablation at $\alpha = 0.01$ over 200 null and 200 post-change streams, with the Wilson 95% interval on the realised PFA. The fourth column reports a checkmark only where the interval's upper end stays at or below nominal, a cross only where its lower end exceeds nominal, and a question mark where neither holds: 0 rows resolve as valid, 5 resolve as invalid, and 14 of 19 are unresolved at 200 replicates — most rows reporting PFA 0.000 fall in this last class, since their Wilson upper bound already exceeds α; see Section 5 for what this design can and cannot certify.

group	variant	PFA [95% CI]	valid?	ARL_0	cens. delay (s)	miss	lag
	SENTINEL-E (full)	0.000 [0.000, 0.019]	?	90,000	244.5	0.01	18
e-process core	changepoint mixture ($\eta = 0$)	0.000 [0.000, 0.019]	?	90,000	311.3	0.06	18
	enable dilation (π grid)	0.000 [0.000, 0.019]	?	90,000	269.8	0.02	18
	single episode length	0.000 [0.000, 0.019]	?	90,000	247.6	0.02	18
	single bet aggressiveness	0.000 [0.000, 0.019]	?	90,000	316.8	0.04	18
temporal	bet on every frame	1.000 [0.981, 1.000]	✗	332	3.3	0.00	1
	bet once per 5 frames	0.935 [0.892, 0.962]	✗	42,640	9.0	0.00	5
	raw calibration (not thinned)	0.010 [0.003, 0.036]	?	89,395	68.4	0.00	18
calibration	pooled conformal	0.975 [0.943, 0.989]	✗	17,415	391.0	0.01	18
	Mondrian conformal	0.025 [0.011, 0.057]	✗	88,848	1737.6	0.54	18
	residual + LR weighting	0.000 [0.000, 0.019]	?	90,000	457.9	0.09	18
conditional validity	DKW inflation	0.000 [0.000, 0.019]	?	90,000	2025.8	0.68	18
	no correction ($\delta = 0$)	0.050 [0.027, 0.090]	✗	88,463	43.1	0.00	18
onset prior	hazard $\rho = 10^{-2}$	0.000 [0.000, 0.019]	?	90,000	563.3	0.10	18
	hazard $\rho = 10^{-4}$	0.000 [0.000, 0.019]	?	90,000	353.0	0.05	18
betting	linear betting (changepoint core)	0.000 [0.000, 0.019]	?	90,000	2597.4	0.92	18
	adaptive ONS betting (changepoint core)	0.000 [0.000, 0.019]	?	90,000	2800.0	1.00	18
	single bet, $K = 1$ (changepoint core)	0.000 [0.000, 0.019]	?	90,000	2775.4	0.99	18
	fixed start (changepoint core)	0.000 [0.000, 0.019]	?	90,000	2800.0	1.00	18

"lag" is the decorrelation lag chosen from the calibration autocorrelation, in frames at 25 fps.

deployment, or that the result transfers to illegal dumping specifically: Avenue and Ped2 are classical pedestrian-walkway anomaly corpora, not dumping footage, because no released corpus — Mivia-IWDD included — provides months of continuously-labelled per-frame dumping-detector output, and the two are not the same task (dumping is committed once and its evidence often persists in the scene; a pedestrian or vehicle anomaly is transient by construction, which is part of why Avenue's episodes are so short). The library ships the splicing protocol and dataset adapters needed to run the identical evaluation on any other corpus with real per-frame scores, and the adapters raise an error rather than substituting synthetic data when those scores are absent; what would close the gap Section 4.2 exposes is not a different evaluation but a backbone whose frame-to-frame errors are more nearly independent than optical-flow magnitude's, which is future work.

The tested horizon is two orders of magnitude short of the motivating claim. The abstract and introduction motivate the problem with a camera running "continuously for weeks" between recalibrations. The longest horizon actually swept in Table 6 is 2.7 hours; a week is 168 hours, roughly 60× longer, and a month roughly 260× longer. Nothing in Theorem 2 or Theorem 1 depends on the horizon — the time-uniform guarantee holds at 10^6 frames exactly as it does at 10^4, and that is the entire content of the word "time-uniform" — so the theoretical claim is unaffected. What is untested is whether the *simulator*'s non-stationarity assumptions, and in particular the diurnal and weather processes that drive the shift studied in Section 4.8, remain plausible representations of real footage at a week's scale rather than a few hours'; slow drifts that do not appear inside 2.7 hours could plausibly compound over a week in ways this experiment cannot show either way. Extending Table 6's sweep by two orders of magnitude is mechanical (the simulator scales to any T) and is one of the two directions we would prioritise in revision.

Conditional super-uniformity is an assumption, not a theorem. Theorem 2 needs the p-values to be super-uniform given the past. Thinning by the decorrelation lag makes the residual dependence small — Table 16 reports the Ljung–Box check [Ljung and Box, 1978] — but it does not eliminate it, and no finite-sample test can certify its absence. Theorem 1 additionally assumes i.i.d. calibration draws, which is why the calibration set is thinned too.

The residual model can be wrong. Equation (1) assumes the normalised residual is context-free. Where it is not, calibration is approximate; the exactly group-conditional Mondrian variant is available for the cases where a single context variable dominates, and the support check bounds the damage from extrapolation but does not detect misspecification inside the support.

Calibration robustness has one measured breaking point, and several unresolved shift levels below it. The residual construction is clearly invalid at $\times 3.0$ shift in an unmodelled covariate (Table 13): the Wilson interval's lower end clears nominal there. At $\times 1.0$, $\times 1.5$, $\times 2.0$ shift the realised rate is not distinguishable from nominal at the replication count run here, which is not the same claim as validity — only Section 4.4's much larger replicate count resolves that question, and it was not re-run at every shift level. Likelihood-ratio weighting holds a realised rate of 0.000 at the one level we can certify as broken for the unweighted construction, but costs most of the power there (miss rate 1.00 against 0.26); we did not verify it holds at the unresolved levels below that, and Section 3.2 discloses that its calibration substitutes an approximation (the Kish effective sample size) for the exact order-statistic argument, which is not itself derived to be valid under likelihood-ratio weights. So the honest summary is that SENTINEL-E's false-alarm guarantee is robust to the shift a well-specified residual model can absorb and to anything the two-sample test catches, and is *not* unconditionally robust: a large shift in a direction the model does not capture, arriving faster than the shift test can react, will inflate the rate, and exactly how large is not established by this experiment beyond the one level that clearly breaks. We know of no distribution-free method for which robustness is unconditional, but that does not make it a non-issue, and an operator should treat a rising abstention or miss rate as a prompt to recalibrate.

The episode model is a model. The two-state chain of Equation (3) assumes episodes with geometric durations and geometric gaps, and a constant within-episode anomalous fraction. Real dumping episodes are none of those exactly. The prior grid spans a wide range of each, and Theorem 2 holds whatever the truth is — the guarantee never depends on the episode model being correct, only the power does. But we have not characterised how power degrades when the true duration distribution is far from anything on the grid, and a heavy-tailed duration distribution is the case we would expect to be worst. The generalisation also costs a little where it is not needed: with a single persistent change the episodic mixture is marginally behind the changepoint mixture (Table 7), because the prior spends mass on episode-end hypotheses that never fire.

Every headline comparison runs at one simulator configuration. Tables 7, 11 and 18 all use 800-frame episodes at 35% intermittency with a 6,000-frame mean gap; Section 4.5's own sweep over episode count and intermittency (Tables 7 and 8) shows the episodic mixture's advantage is not uniform across that grid — it is absent or reversed at one episode and at low intermittency (Section 3.3.2) — so a reader should not extrapolate the headline numbers to a configuration outside the swept range. We have not run the full baseline comparison or the ablation at multiple points in $(\text{episode length}, \text{gap}, \pi)$-space, only at the two axes Section 4.5 varies one at a time; a joint sweep is mechanical but was not run here.

Fleet scale and graph knowledge. Experiments use 12 cameras with a known planar geometry. Larger fleets are computationally trivial (Table 15), but we have not tested whether the learned prior transfers across topologies, and a municipality whose camera positions are poorly recorded would need to infer the graph.

An operator cannot, from the alarm stream alone, tell a working camera from a silently conservative one. A camera whose calibration set has drifted out of support abstains and bets neutrally (Section 3.2); one whose realised miss rate has risen because of an unmodelled shift produces *fewer* alarms, not more, and looks identical from outside to a camera at a genuinely quiet site. The miss rate itself is unobservable in the field, by definition — it is a count over events that were never flagged — and the one instrument that could catch a silent failure, the two-sample shift test of Section 3.2, is off unless likelihood-ratio weighting is explicitly enabled, which is not the default path reported in Tables 5 and 11. We do not have a fix for this in the present work; a deployed system should expose the abstention rate, the shift-test statistic, and the calibration set's age as first-class operator-facing signals rather than internal diagnostics, and should not default the shift test off.

Deployment guidance this paper does not give. Nothing here tells an operator how to choose α, how large a calibration set to collect before trusting the alarm rate, or when a rising abstention or miss rate should trigger recalibration rather than a shrug. Section 4.4 fixes α and studies its consequences; it does not study how a municipality should set it against the cost of a false dispatch and the cost of a missed event, which is a decision-theoretic question this paper leaves open. Relevant literatures we have not engaged include alarm management and human factors (how many alerts per shift a human operator can act on before alarm fatigue sets in), classical signal detection theory (the ROC-operating-point framing a control-room already uses for other sensors), statistical process control practice (where a control chart's recalibration cadence is itself a studied design choice), and public-sector algorithmic accountability (what disclosure a municipality owes residents about an automated flagging system run against public spaces). Each bears directly on how this method would actually be operated, and none of it is in Section 2.

Governance and human oversight. This is a detector, not a decision-maker, and the gap between those two is where the harder questions live. An alarm is a prompt for a human to look at footage and decide whether dumping occurred and who is responsible; the false-alarm guarantee in this paper says nothing about what happens after that prompt — whether the human reviewing it is adequately resourced, whether the review is itself audited, or whether the system is used to justify automated enforcement action without one. Camera placement, retention policy, and who may query the alarm history are governance decisions this paper does not make and should not be read as endorsing any particular answer to. A municipality deploying this or a similar system should treat the statistical guarantee as one input to a decision that also needs an explicit data-retention policy, a named human reviewer in the loop before any enforcement action, and a mechanism by which a resident can contest a flagged event — none of which follows from, or is addressed by, the anytime-valid construction itself.

6 Conclusion

Waste-surveillance systems are evaluated per clip and deployed per month, and the statistical consequence of that mismatch is larger than the field's reporting practice reveals: valid per-frame p-values, applied with the obvious stopping rule, produce a false alarm on essentially every multi-hour null stream we simulated. The 2026 IWDD contest, drawing ten teams, reports per-clip F_1 plus a notification-delay measure and, in its own overview, a per-clip ROC analysis; none of that reports a rate over a continuous monitoring horizon.

Closing *that* gap does not need the episodic construction: any finite-sample anytime-valid e-process — including the changepoint mixture this paper takes as its own point of departure, which by Proposition 4 is a faithful instance of prior work [Saha and Ramdas, 2026] — composed with the same conformal front end already holds the nominal level in our own experiments (Section 4.12). What required the new construction is a narrower, and still real, claim: a dumping event is an episode that ends and recurs, and mixing over episodes rather than changepoint locations buys back detection power in that regime without touching the guarantee. The episodic e-process mixes over when an episode starts *and stops*, remains an exact supermartingale so Ville's inequality applies unchanged, costs $O(1)$ per frame, and contains the changepoint mixture exactly as the boundary case $\eta = 0$. Around it, conformal calibration conditional on observable context makes the p-values honest, and e-BH turns per-camera evidence into fleet-wide decisions under arbitrary dependence. Three components did not survive contact with measurement and are reported rather than removed: the dilated emission for within-episode intermittency, which the betting mixture already handles; the learned spatial prior, which leaves the guarantee exactly intact — as its theorem promises — while failing to improve detection; and the size, and even the sign, of the episodic mixture's power advantage over the changepoint mixture, which Section 4.5 resolves with a paired analysis on matched seeds and a shared betting grid rather than the unpaired comparison an earlier, confounded version of that experiment reported. Both mixtures are, in our comparison, among the methods that hold their nominal false-alarm level; SENTINEL-E is the only one of the two that keeps the episode-recurrence axis available at the same $O(1)$ cost, at 0.0016% of the cost of the backbone it wraps — with a measured boundary, reported rather than elided, beyond which a shift in an unmodelled covariate breaks it unless the conservative safeguard is engaged.

This revision closes part of the first empirical direction the earlier version of this paper deferred: Section 4.2 runs the identical pipeline, unmodified, on real per-frame scores from two real surveillance corpora, CUHK Avenue and UCSD Ped2, rather than on simulated streams alone. What it finds is not unqualified success. The guarantee holds on both real corpora at the lag each dataset's own diagnostic selects, and every classical baseline tested fails to hold it on the same real streams — but the window in which validity and power coexist at all is narrow, narrower than any simulated configuration in this paper, because real optical-flow scores are far more autocorrelated than an AR(1) process and real anomalous episodes are correspondingly brief. Two directions follow from that finding specifically. The nearer one is a stronger backbone: what Section 4.2 exposes is a property of the scores, not of the sequential layer, and a backbone whose frame-to-frame errors are more nearly independent than optical-flow magnitude's would widen the window without touching the theory. The further one is the corpus itself: this paper's real-data validation runs on pedestrian-walkway anomaly footage because no continuously-labelled real illegal-dumping stream of the required length is publicly available, and closing that gap needs a released corpus, from Mivia-IWDD or an operational municipal deployment, rather than a change to the method. Neither the argument nor the algorithm is specific to dumping in the first place. Episodic, intermittent, recurring events observed through a continuously-monitored sensor describe wildlife poaching, illegal fishing, gas leaks and infrastructure faults as well as the loitering, wrong-direction motion and abandoned-object anomalies the real corpora above already represent; wherever a field evaluates on clips and deploys on streams, the same optional-stopping problem and the same remedy apply.

Declarations

Funding. No funding was received for this work.

Competing interests. The author declares no competing interests.

Data and code availability. All code, experiment scripts and generated results are available in the project repository. The corpora referenced in Section 2 are distributed by their respective authors under their own terms; no data from them is redistributed here.

Ethical approval. Not applicable. No human subjects data was used; all experiments run on simulated streams. This work concerns the statistical calibration of alerting systems, not the governance of their deployment; Section 5 gives a substantive treatment of the human-oversight, accountability, and disclosure questions that automated surveillance of public space raises, rather than treating the topic as out of scope.

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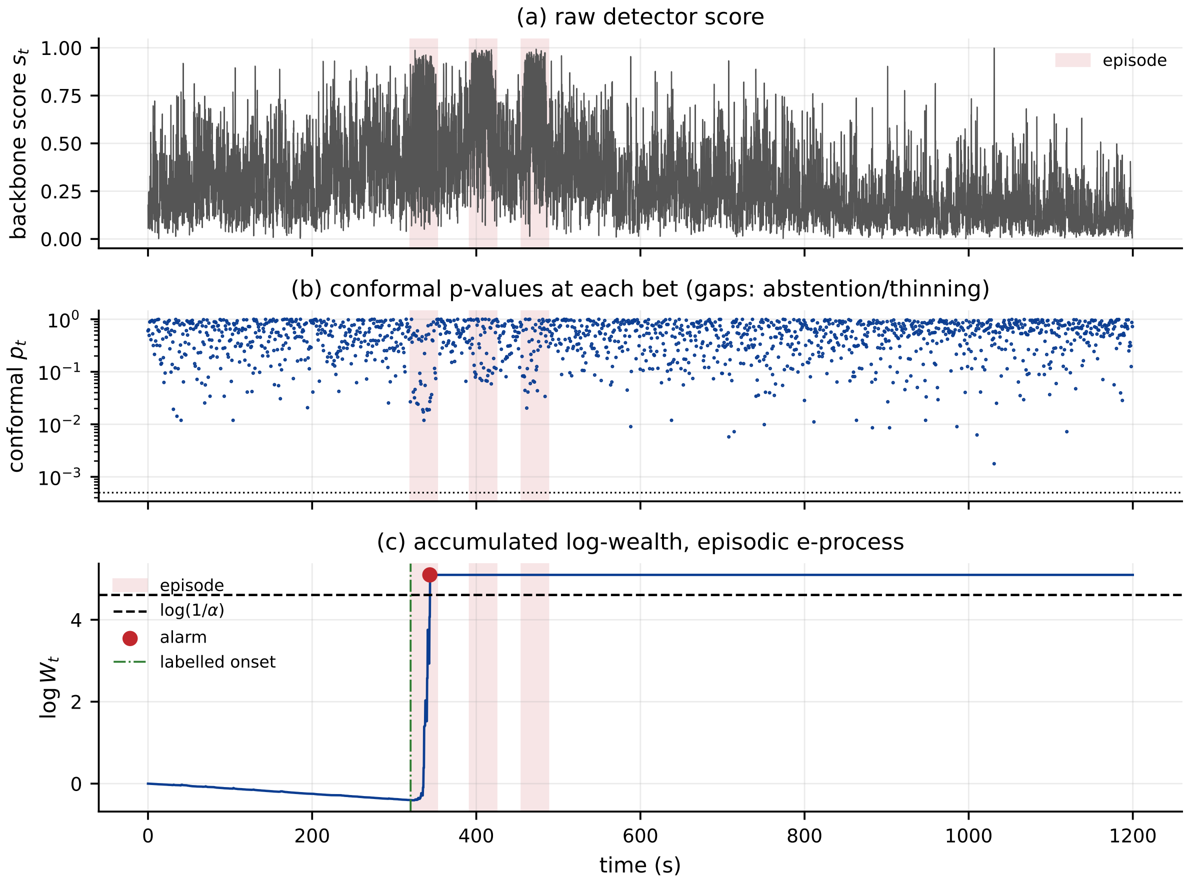

Figure 3: One simulated 1200-second stream with 3 episodes (shaded), chosen for its episode count rather than for how favourably it behaves. **(a)** the raw backbone score, spiking during each episode but also, by construction, during ordinary null activity. **(b)** the conformal p-value at each betting instant (log scale); gaps are frames the decorrelation lag skips or the calibration abstains on. Notice that the score spikes in (a) do not translate into uniformly small p-values in (b) — the score distribution the conformal layer calibrates against already absorbs much of the diurnal and weather-driven variation, so only the residual that context cannot explain shows up as evidence. **(c)** the episodic e-process's accumulated log-wealth. The dash-dot line is the labelled onset; the process is already drifting down slightly beforehand (an ordinary null fluctuation, licensed by the supermartingale property), climbs sharply once the first episode's evidence accumulates, and crosses $\log(1/\alpha)$ 23.4 s after onset — the red point.

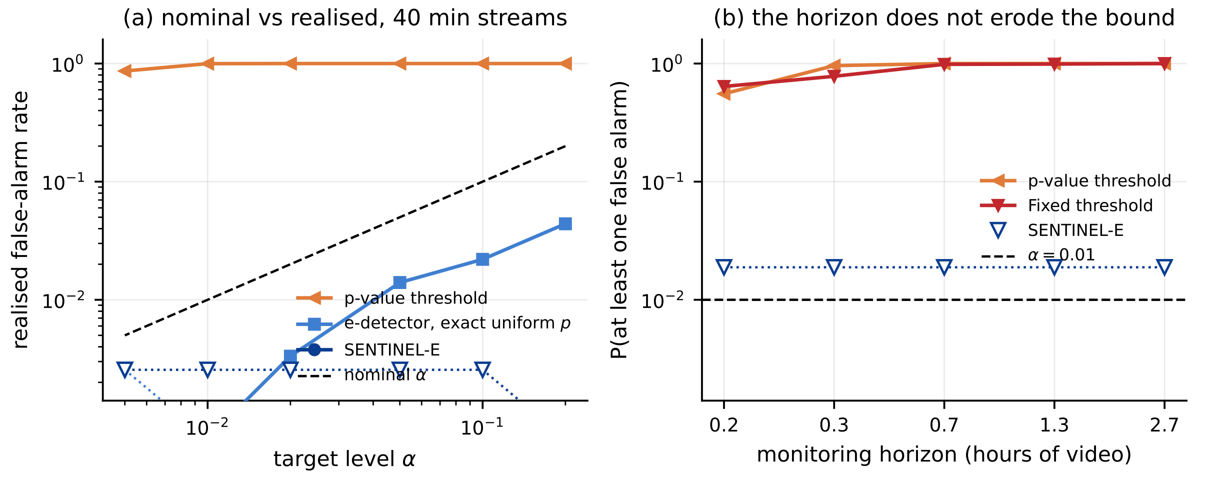

Figure 4: Time-uniform false-alarm control, 1500 null streams per point in (a) and 200 in (b). Open downward triangles mark operating points at which *no* false alarm was observed; they are drawn at the 95% Wilson upper confidence limit, which is the strongest statement the sample supports, and the dotted line connects them. **(a)** Realised against nominal level. SENTINEL-E stays below the diagonal everywhere; the per-frame p-value threshold, fed *identical* conformal p-values, sits at the top of the plot, so the difference is attributable to the stopping rule alone. The middle curve is the same e-detector fed exactly uniform p-values, which separates the slack in Ville's inequality from the conservatism of the conformal layer. **(b)** Against the monitoring horizon at $\alpha = 0.01$. The per-frame rules approach one; SENTINEL-E is flat.

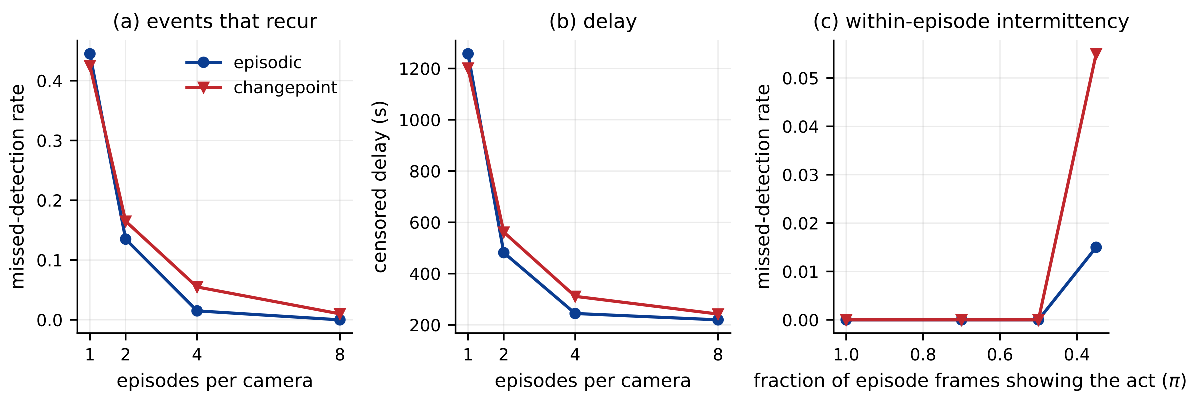

Figure 5: What mixing over episodes buys, and where it does not, on the unpaired point estimates (Table 7). With a single episode the episodic mixture is nominally behind rather than tied with the changepoint mixture — Table 9 resolves this as a real, if small, cost; the paired analysis below is what head-to-head claims in this section rest on.

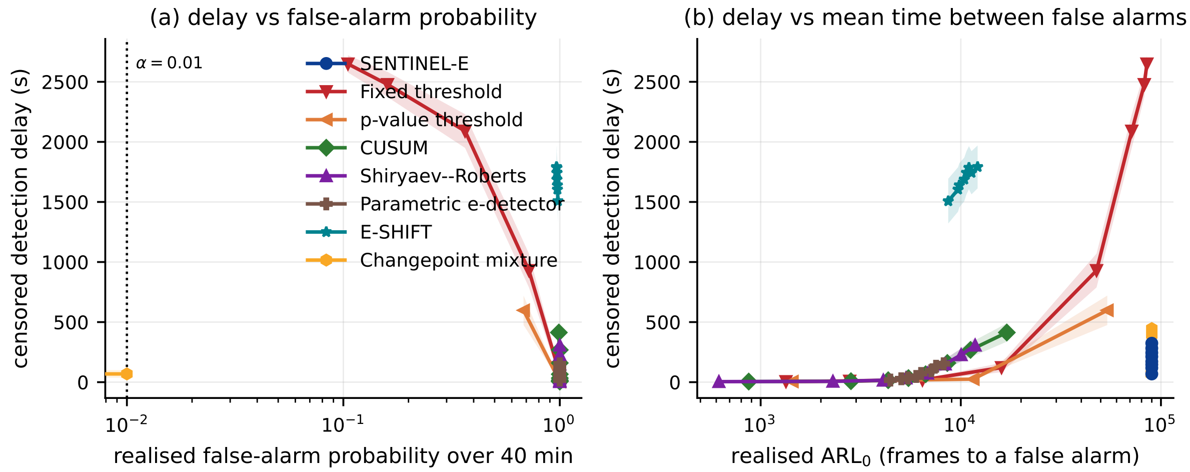

Figure 6: Detection delay against realised false-alarm rate, 200 null and 200 post-change streams per operating point. Lower and further left is better; the operating region a municipality can live with is the left-hand edge. Each method is swept over its own knob and plotted where it actually lands, not where it claims to.

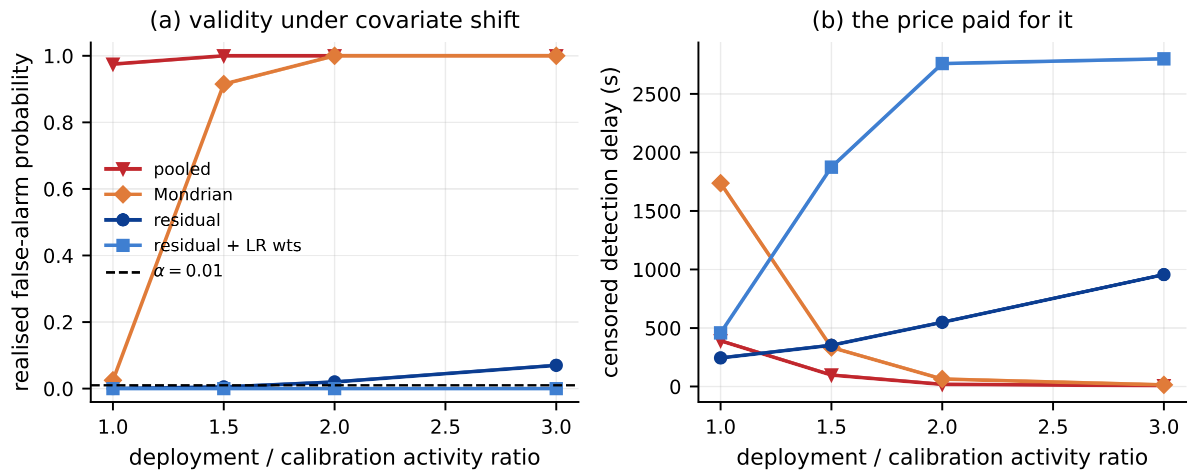

Figure 7: Shift in an *unbinned* covariate (scene activity). Binning the context cannot help with a variable outside the taxonomy, so Mondrian calibration fails alongside the pooled baseline; conformalising the context-normalised residual holds throughout.

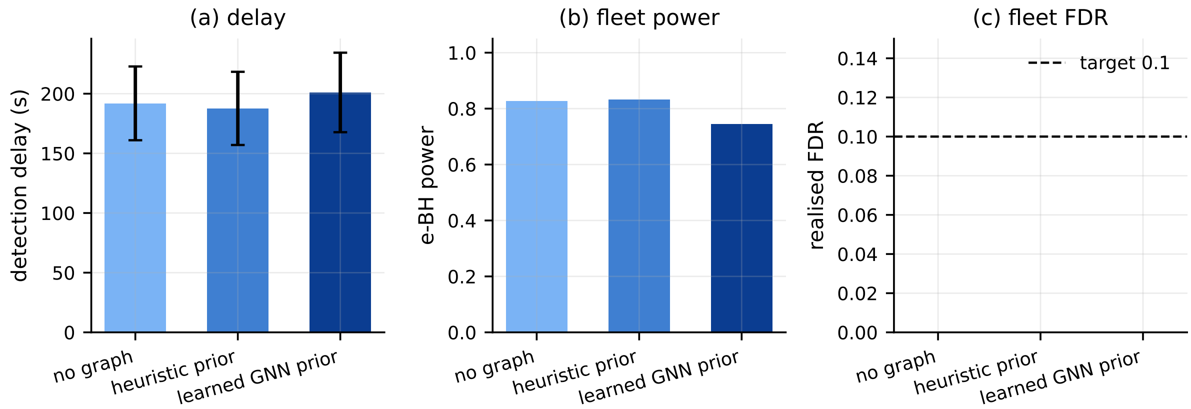

Figure 8: Fleet of 12 cameras over 40 held-out fleets with events and 40 all-null fleets. The learned spatial prior does *not* shorten delay — it is worse than no graph at all, the negative result detailed below — while e-BH holds the false-discovery rate under the dependence the graph induces regardless of which controller feeds it.